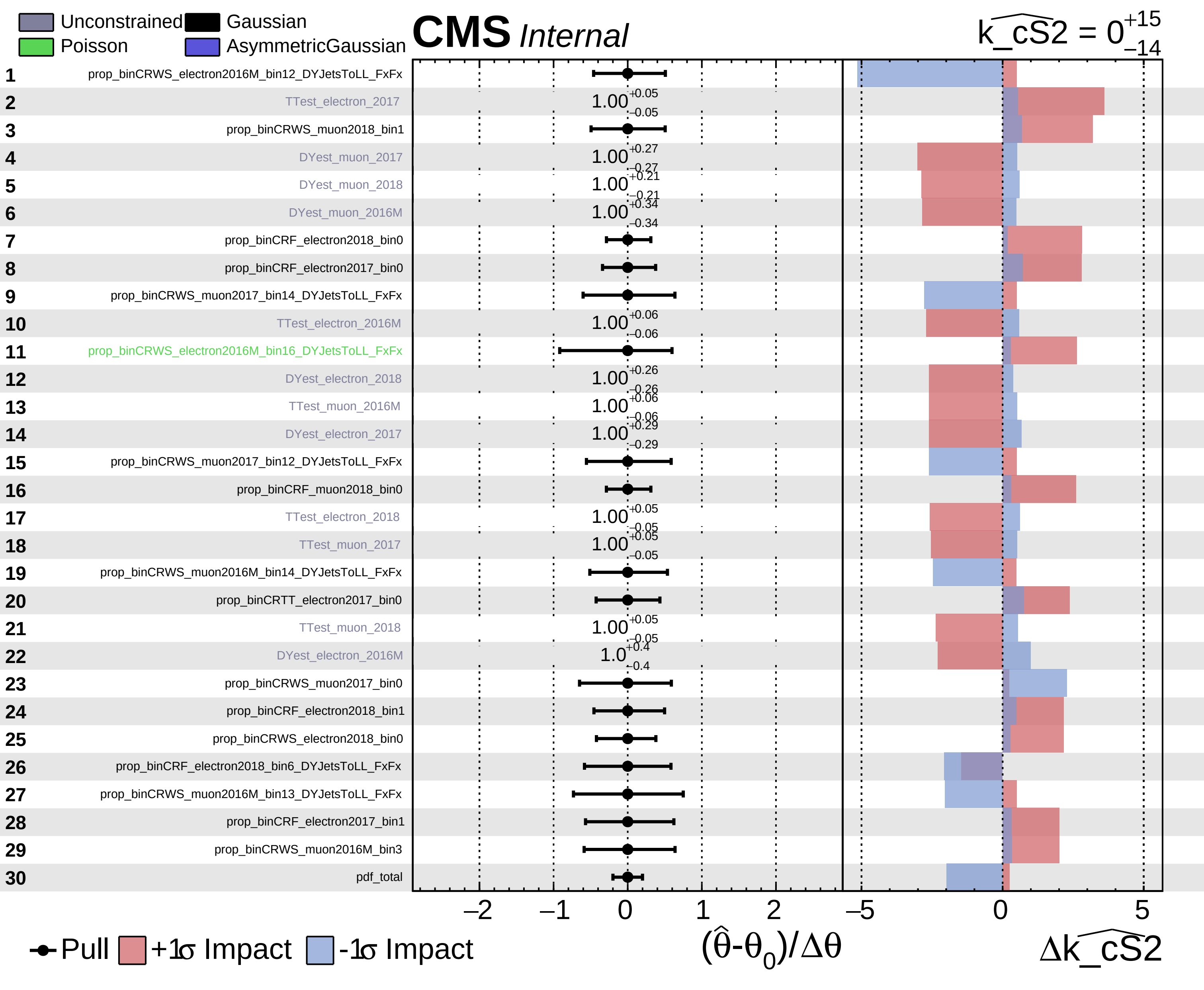
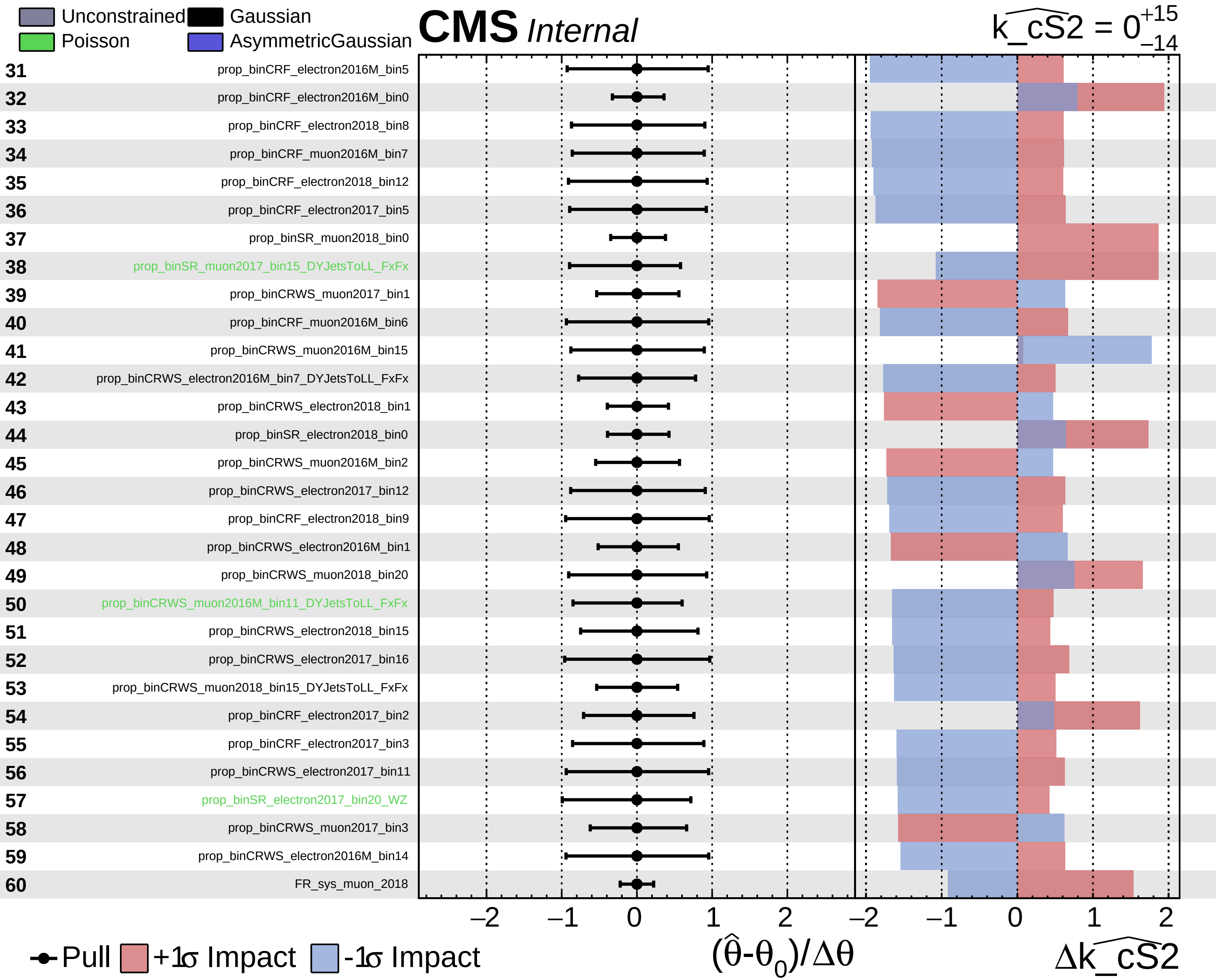
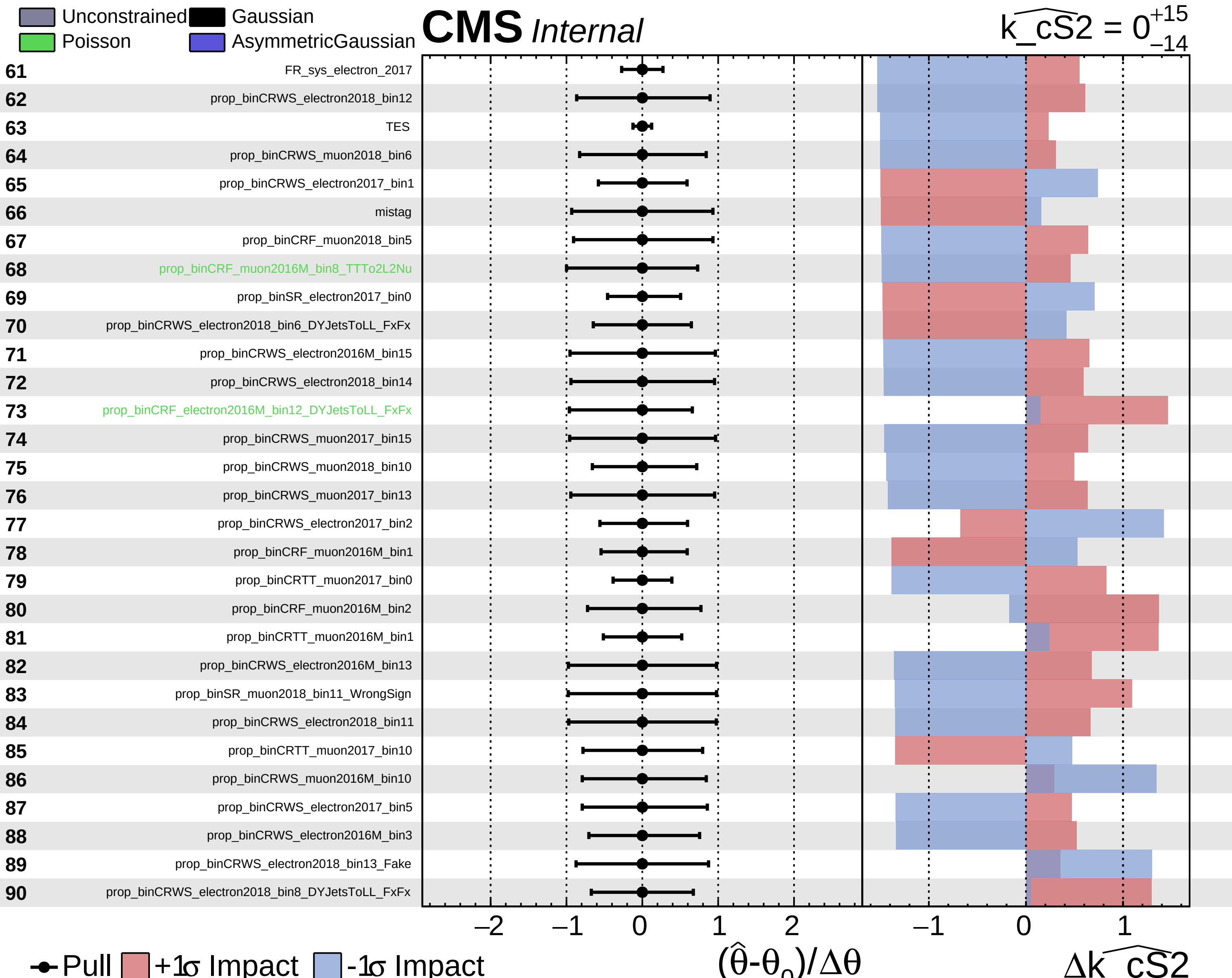
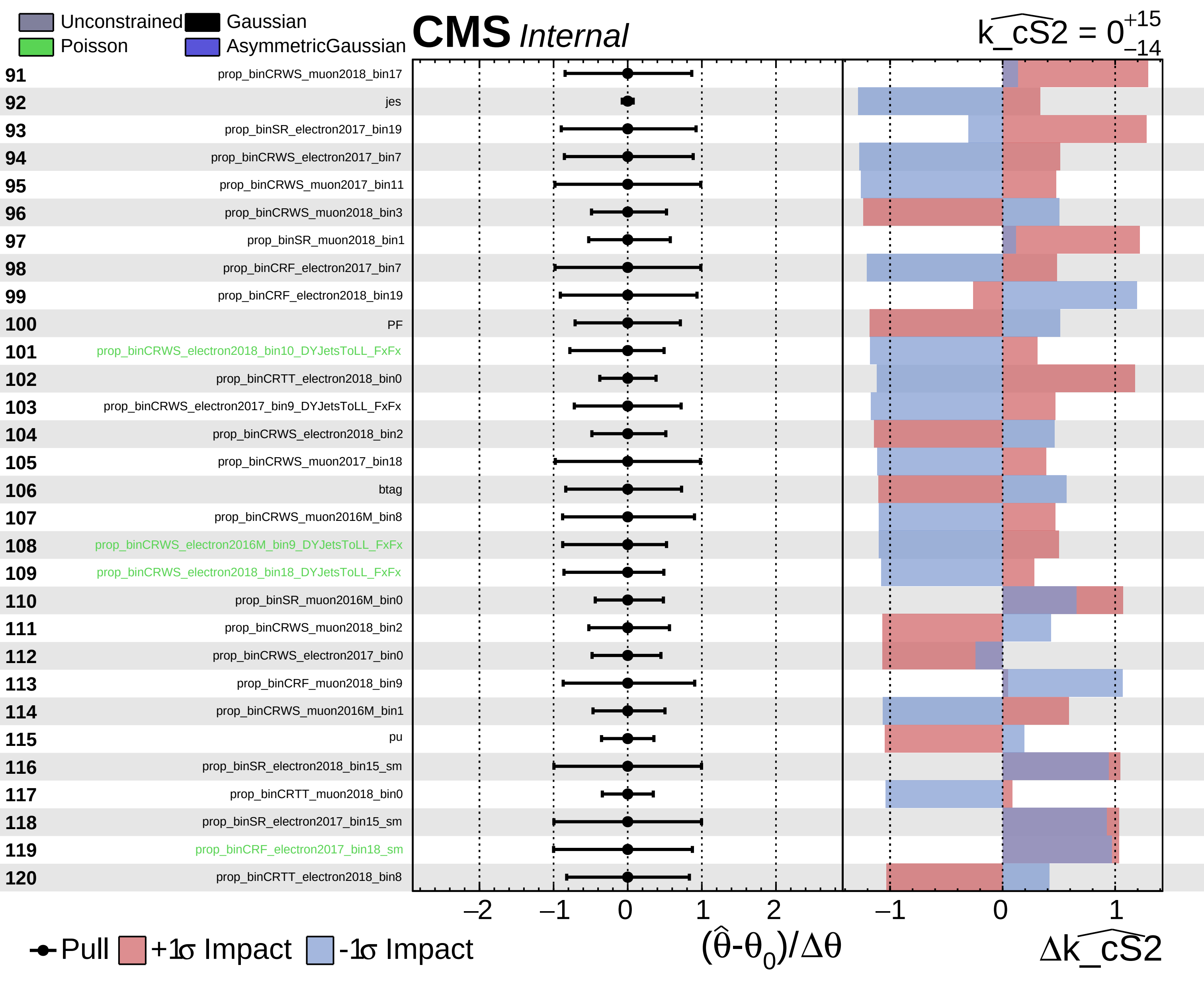


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
1 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
2 TTest_electron_2017 1.00 +0.05 −0.05
3 prop_binCRWS_muon2018_bin1
4 DYest_muon_2017 1.00 +0.27 −0.27
5 DYest_muon_2018 1.00 +0.21 −0.21
6 DYest_muon_2016M 1.00 +0.34 −0.34
7 prop_binCRF_electron2018_bin0
8 prop_binCRF_electron2017_bin0
9 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
10 TTest_electron_2016M 1.00 +0.06 −0.06
11 prop_binCRWS_electron2016M_bin16_DYJetsToLL_FxFx
12 DYest_electron_2018 1.00 +0.26 −0.26
13 TTest_muon_2016M 1.00 +0.06 −0.06
14 DYest_electron_2017 1.00 +0.29 −0.29
15 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
16 prop_binCRF_muon2018_bin0
17 TTest_electron_2018 1.00 +0.05 −0.05
18 TTest_muon_2017 1.00 +0.05 −0.05
19 prop_binCRWS_muon2016M_bin14_DYJetsToLL_FxFx
20 prop_binCRTT_electron2017_bin0
21 TTest_muon_2018 1.00 +0.05 −0.05
22 DYest_electron_2016M 1.0 +0.4 −0.4
23 prop_binCRWS_muon2017_bin0
24 prop_binCRF_electron2018_bin1
25 prop_binCRWS_electron2018_bin0
26 prop_binCRF_electron2018_bin6_DYJetsToLL_FxFx
27 prop_binCRWS_muon2016M_bin13_DYJetsToLL_FxFx
28 prop_binCRF_electron2017_bin1
29 prop_binCRWS_muon2016M_bin3
30 pdf_total
−2 −1 0 1 2
−5 0 5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2

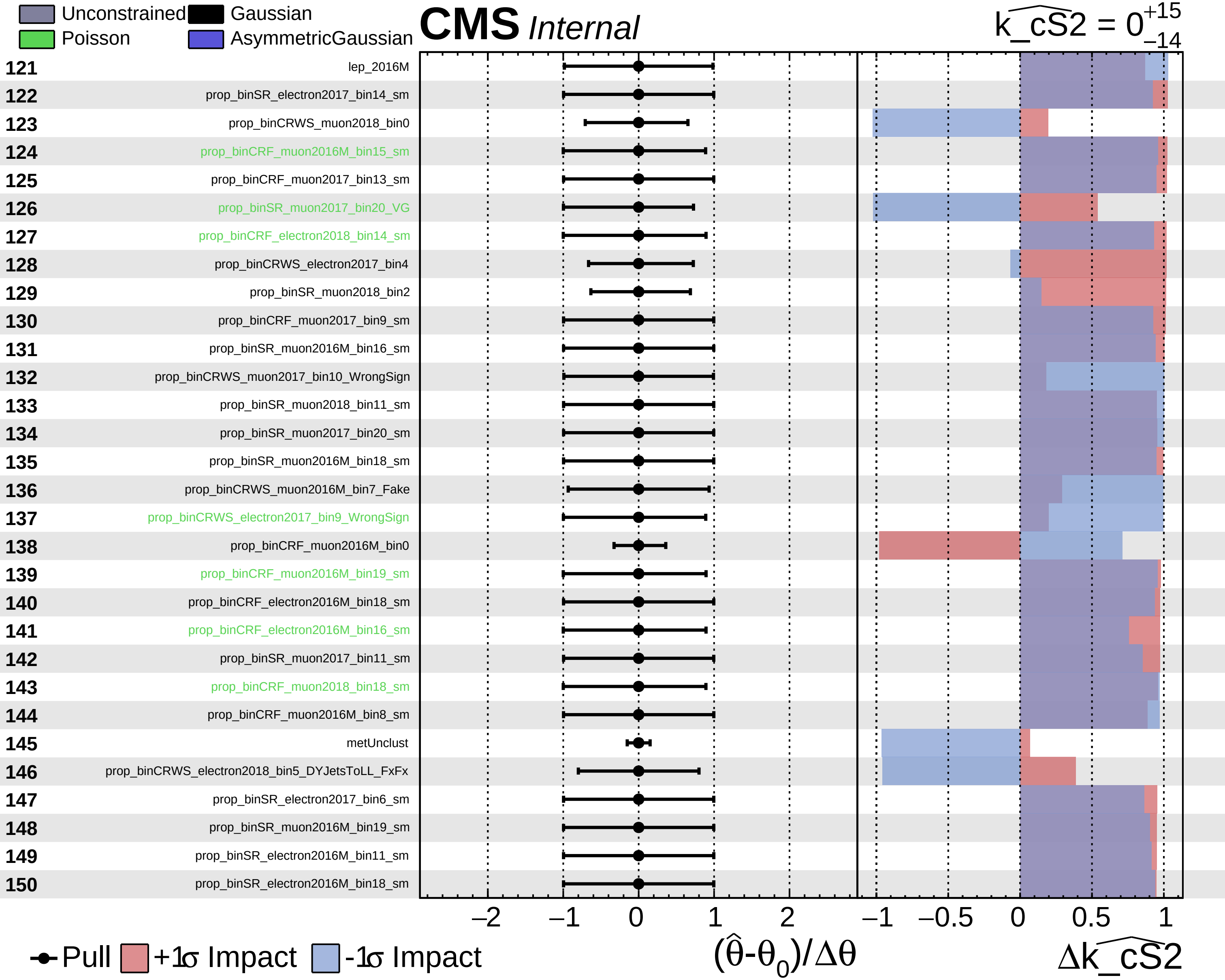

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
31 prop_binCRF_electron2016M_bin5
32 prop_binCRF_electron2016M_bin0
33 prop_binCRF_electron2018_bin8
34 prop_binCRF_muon2016M_bin7
35 prop_binCRF_electron2018_bin12
36 prop_binCRF_electron2017_bin5
37 prop_binSR_muon2018_bin0
38 prop_binSR_muon2017_bin15_DYJetsToLL_FxFx
39 prop_binCRWS_muon2017_bin1
40 prop_binCRF_muon2016M_bin6
41 prop_binCRWS_muon2016M_bin15
42 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
43 prop_binCRWS_electron2018_bin1
44 prop_binSR_electron2018_bin0
45 prop_binCRWS_muon2016M_bin2
46 prop_binCRWS_electron2017_bin12
47 prop_binCRF_electron2018_bin9
48 prop_binCRWS_electron2016M_bin1
49 prop_binCRWS_muon2018_bin20
50 prop_binCRWS_muon2016M_bin11_DYJetsToLL_FxFx
51 prop_binCRWS_electron2018_bin15
52 prop_binCRWS_electron2017_bin16
53 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
54 prop_binCRF_electron2017_bin2
55 prop_binCRF_electron2017_bin3
56 prop_binCRWS_electron2017_bin11
57 prop_binSR_electron2017_bin20_WZ
58 prop_binCRWS_muon2017_bin3
59 prop_binCRWS_electron2016M_bin14
60 FR_sys_muon_2018
−2 −1 0 1 2
−2 −1 0 1 2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2



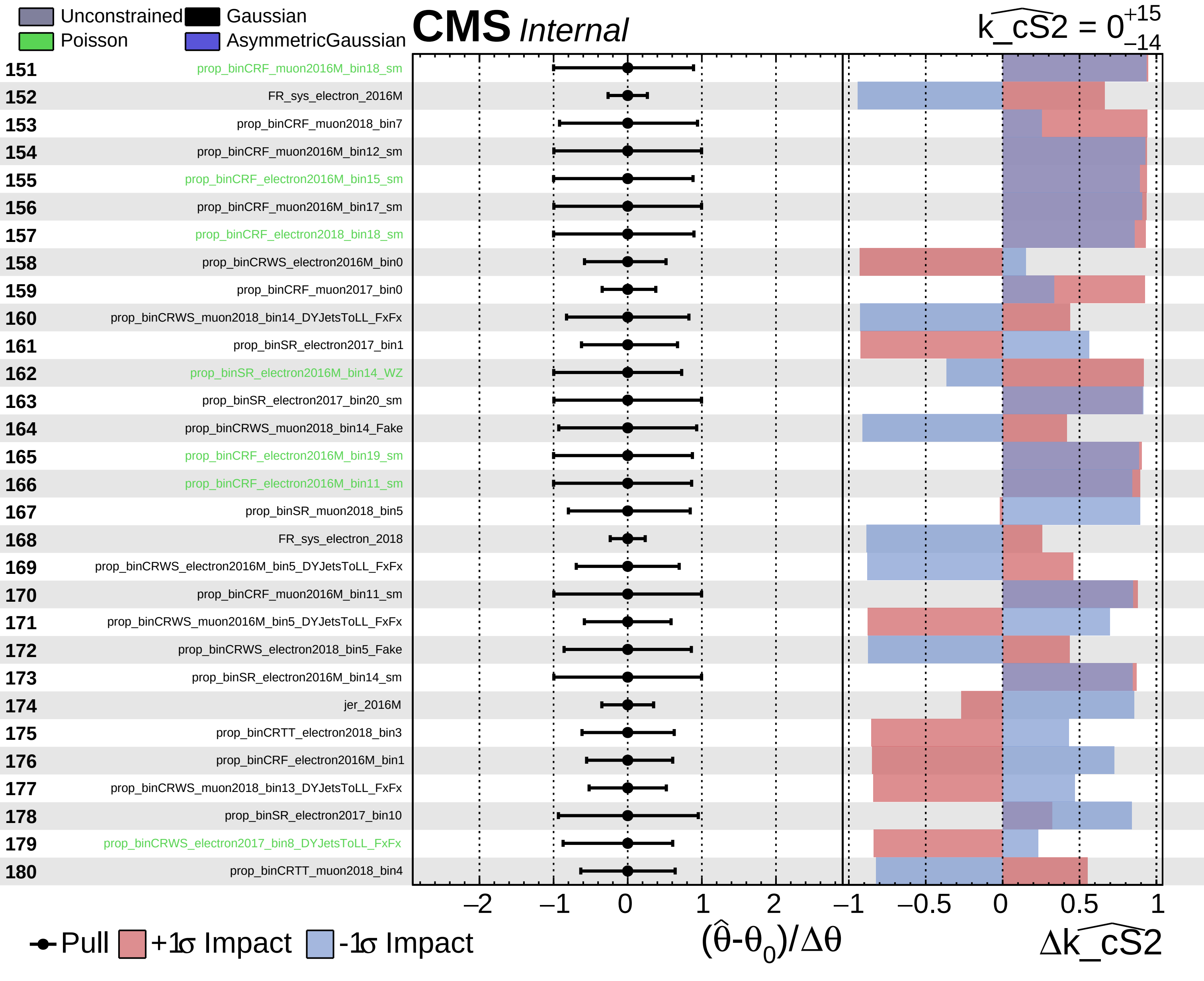
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0^{+15}_{-14}
61 FR_sys_electron_2017
62 prop_binCRWS_electron2018_bin12
63 TES
64 prop_binCRWS_muon2018_bin6
65 prop_binCRWS_electron2017_bin1
66 mistag
67 prop_binCRF_muon2018_bin5
68 prop_binCRF_muon2016M_bin8_TTTo2L2Nu
69 prop_binSR_electron2017_bin0
70 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
71 prop_binCRWS_electron2016M_bin15
72 prop_binCRWS_electron2018_bin14
73 prop_binCRF_electron2016M_bin12_DYJetsToLL_FxFx
74 prop_binCRWS_muon2017_bin15
75 prop_binCRWS_muon2018_bin10
76 prop_binCRWS_muon2017_bin13
77 prop_binCRWS_electron2017_bin2
78 prop_binCRF_muon2016M_bin1
79 prop_binCRTT_muon2017_bin0
80 prop_binCRF_muon2016M_bin2
81 prop_binCRTT_muon2016M_bin1
82 prop_binCRWS_electron2016M_bin13
83 prop_binSR_muon2018_bin11_WrongSign
84 prop_binCRWS_electron2018_bin11
85 prop_binCRTT_muon2017_bin10
86 prop_binCRWS_muon2016M_bin10
87 prop_binCRWS_electron2017_bin5
88 prop_binCRWS_electron2016M_bin3
89 prop_binCRWS_electron2018_bin13_Fake
90 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
−2
−1
0
1
2
−1
0
1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta \widehat{k\_cS2}$



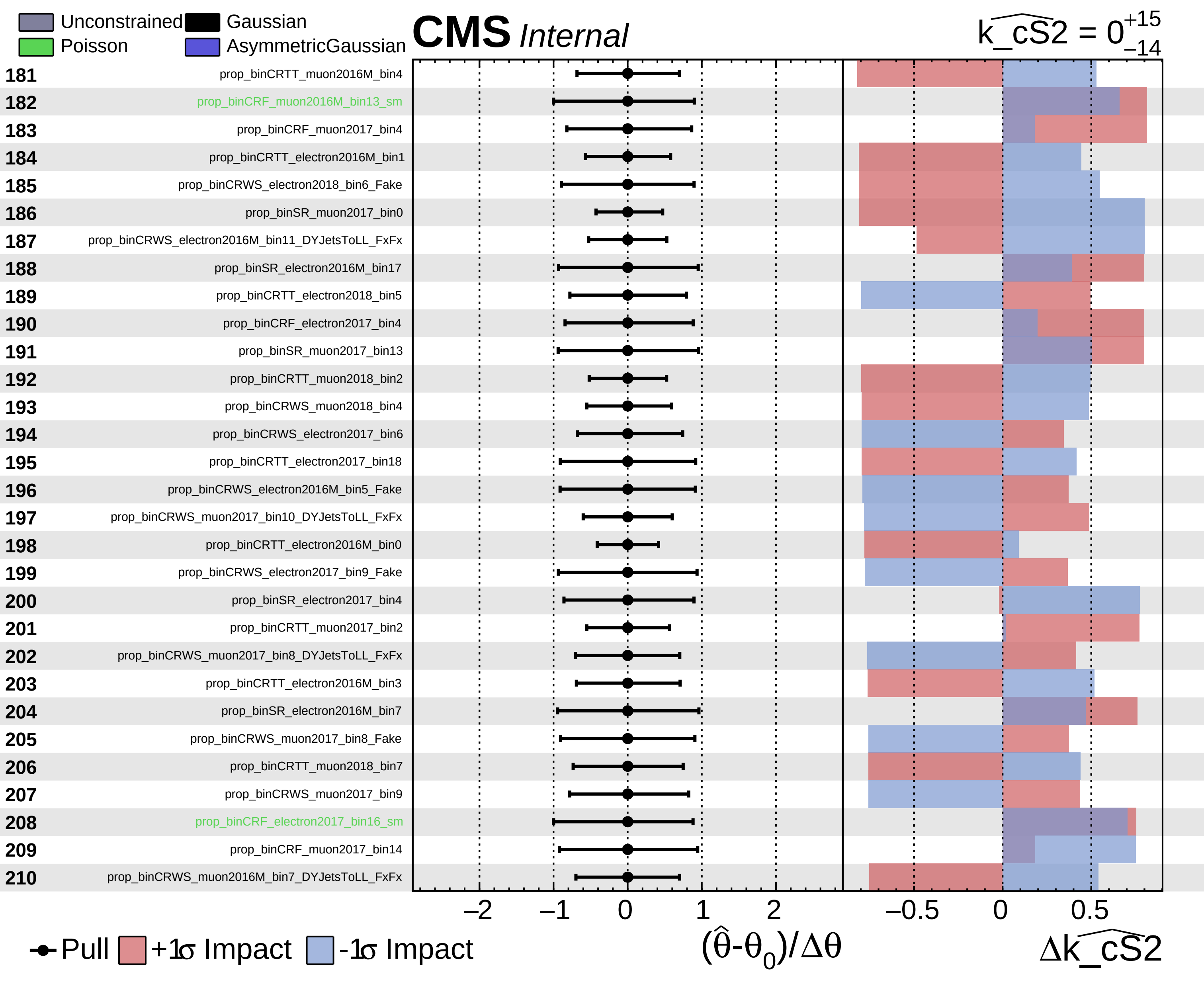
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
91 prop_binCRWS_muon2018_bin17
92 jes
93 prop_binSR_electron2017_bin19
94 prop_binCRWS_electron2017_bin7
95 prop_binCRWS_muon2017_bin11
96 prop_binCRWS_muon2018_bin3
97 prop_binSR_muon2018_bin1
98 prop_binCRF_electron2017_bin7
99 prop_binCRF_electron2018_bin19
100 PF
101 prop_binCRWS_electron2018_bin10_DYJetsToLL_FxFx
102 prop_binCRTT_electron2018_bin0
103 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
104 prop_binCRWS_electron2018_bin2
105 prop_binCRWS_muon2017_bin18
106 btag
107 prop_binCRWS_muon2016M_bin8
108 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
109 prop_binCRWS_electron2018_bin18_DYJetsToLL_FxFx
110 prop_binSR_muon2016M_bin0
111 prop_binCRWS_muon2018_bin2
112 prop_binCRWS_electron2017_bin0
113 prop_binCRF_muon2018_bin9
114 prop_binCRWS_muon2016M_bin1
115 pu
116 prop_binSR_electron2018_bin15_sm
117 prop_binCRTT_muon2018_bin0
118 prop_binSR_electron2017_bin15_sm
119 prop_binCRF_electron2017_bin18_sm
120 prop_binCRTT_electron2018_bin8
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



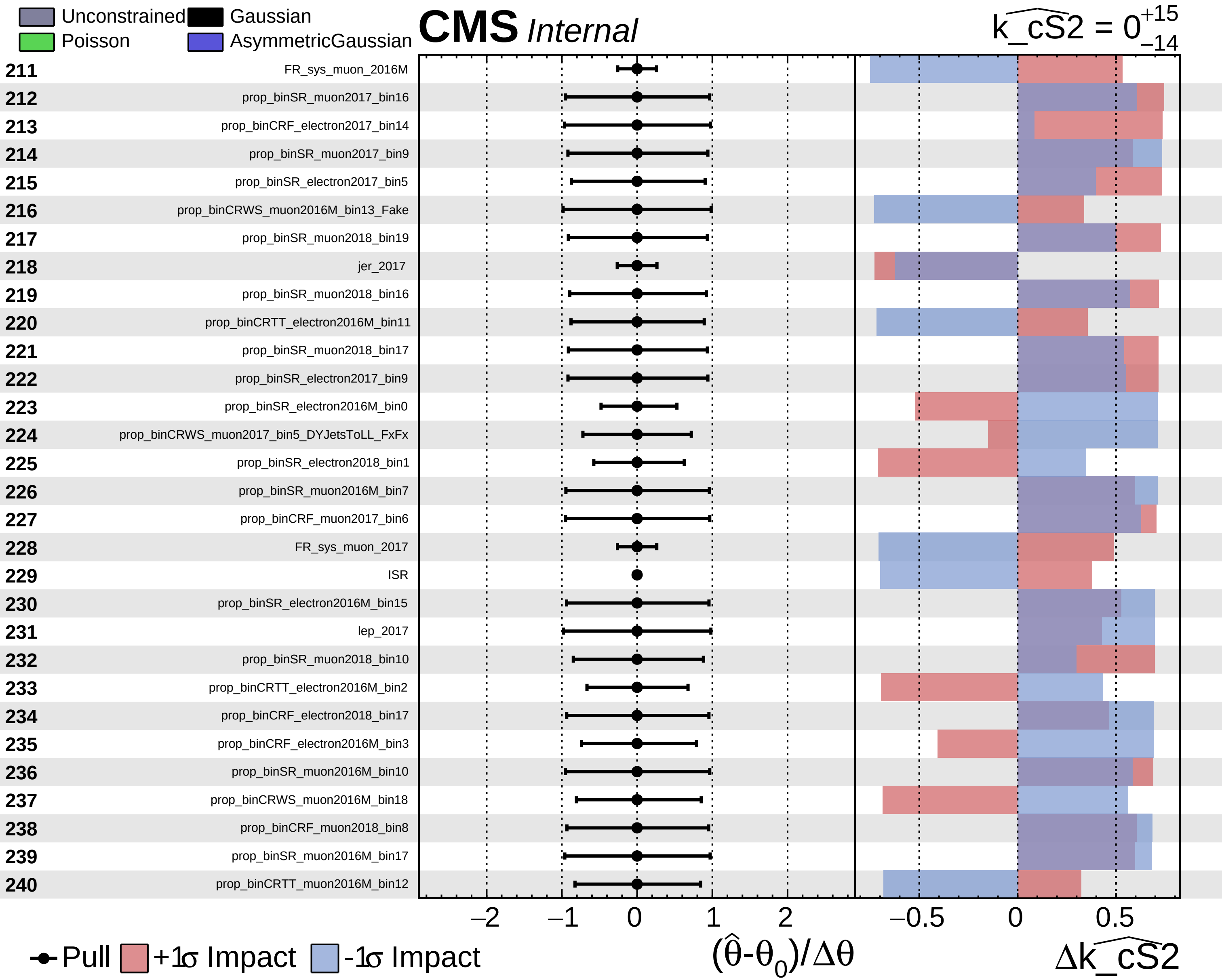
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
121 lep_2016M
122 prop_binSR_electron2017_bin14_sm
123 prop_binCRWS_muon2018_bin0
124 prop_binCRF_muon2016M_bin15_sm
125 prop_binCRF_muon2017_bin13_sm
126 prop_binSR_muon2017_bin20_VG
127 prop_binCRF_electron2018_bin14_sm
128 prop_binCRWS_electron2017_bin4
129 prop_binSR_muon2018_bin2
130 prop_binCRF_muon2017_bin9_sm
131 prop_binSR_muon2016M_bin16_sm
132 prop_binCRWS_muon2017_bin10_WrongSign
133 prop_binSR_muon2018_bin11_sm
134 prop_binSR_muon2017_bin20_sm
135 prop_binSR_muon2016M_bin18_sm
136 prop_binCRWS_muon2016M_bin7_Fake
137 prop_binCRWS_electron2017_bin9_WrongSign
138 prop_binCRF_muon2016M_bin0
139 prop_binCRF_muon2016M_bin19_sm
140 prop_binCRF_electron2016M_bin18_sm
141 prop_binCRF_electron2016M_bin16_sm
142 prop_binSR_muon2017_bin11_sm
143 prop_binCRF_muon2018_bin18_sm
144 prop_binCRF_muon2016M_bin8_sm
145 metUnclust
146 prop_binCRWS_electron2018_bin5_DYJetsToLL_FxFx
147 prop_binSR_electron2017_bin6_sm
148 prop_binSR_muon2016M_bin19_sm
149 prop_binSR_electron2016M_bin11_sm
150 prop_binSR_electron2016M_bin18_sm
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



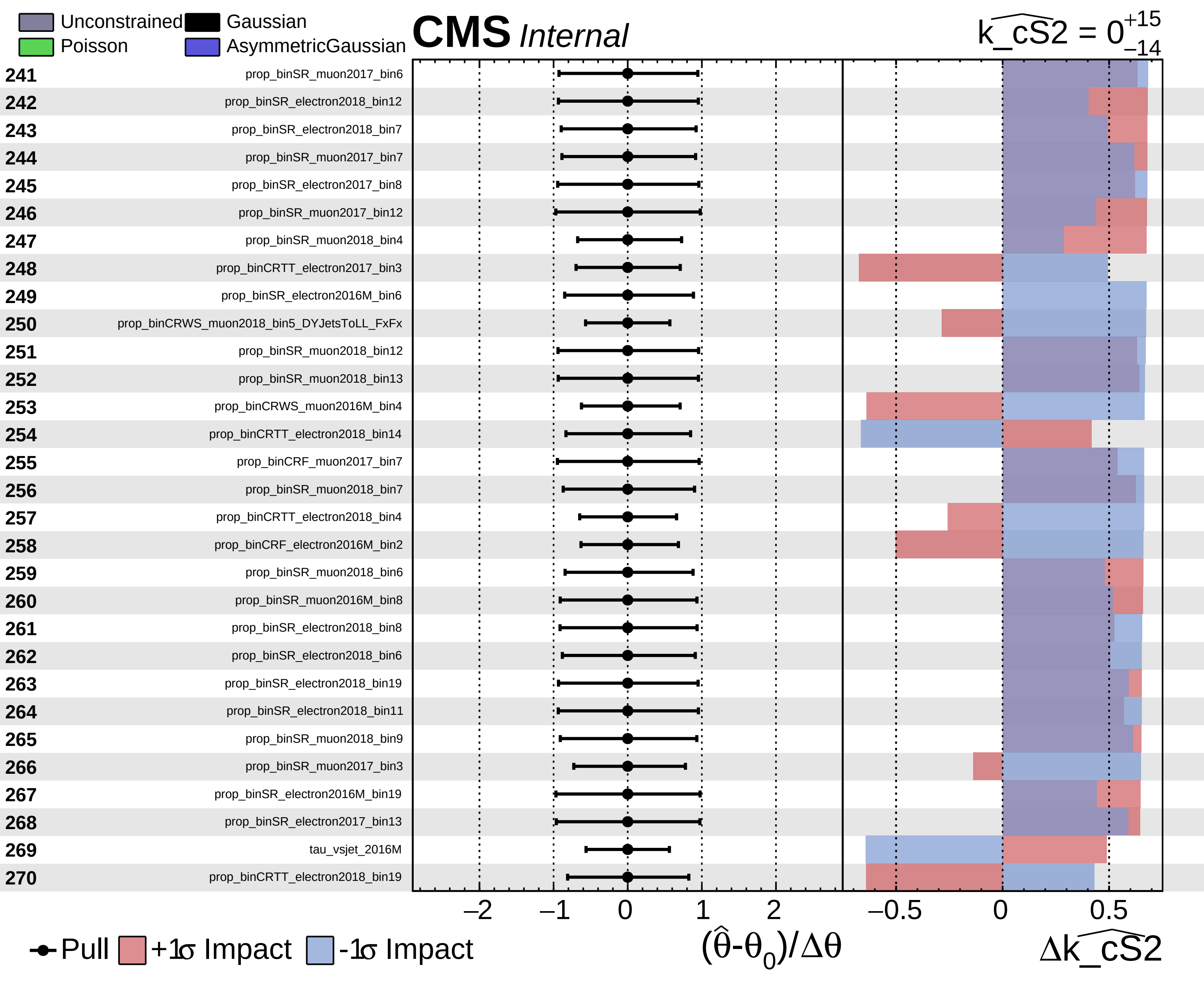
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
151 prop_binCRF_muon2016M_bin18_sm
152 FR_sys_electron_2016M
153 prop_binCRF_muon2018_bin7
154 prop_binCRF_muon2016M_bin12_sm
155 prop_binCRF_electron2016M_bin15_sm
156 prop_binCRF_muon2016M_bin17_sm
157 prop_binCRF_electron2018_bin18_sm
158 prop_binCRWS_electron2016M_bin0
159 prop_binCRF_muon2017_bin0
160 prop_binCRWS_muon2018_bin14_DYJetsToLL_FxFx
161 prop_binSR_electron2017_bin1
162 prop_binSR_electron2016M_bin14_WZ
163 prop_binSR_electron2017_bin20_sm
164 prop_binCRWS_muon2018_bin14_Fake
165 prop_binCRF_electron2016M_bin19_sm
166 prop_binCRF_electron2016M_bin11_sm
167 prop_binSR_muon2018_bin5
168 FR_sys_electron_2018
169 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
170 prop_binCRF_muon2016M_bin11_sm
171 prop_binCRWS_muon2016M_bin5_DYJetsToLL_FxFx
172 prop_binCRWS_electron2018_bin5_Fake
173 prop_binSR_electron2016M_bin14_sm
174 jer_2016M
175 prop_binCRTT_electron2018_bin3
176 prop_binCRF_electron2016M_bin1
177 prop_binCRWS_muon2018_bin13_DYJetsToLL_FxFx
178 prop_binSR_electron2017_bin10
179 prop_binCRWS_electron2017_bin8_DYJetsToLL_FxFx
180 prop_binCRTT_muon2018_bin4
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



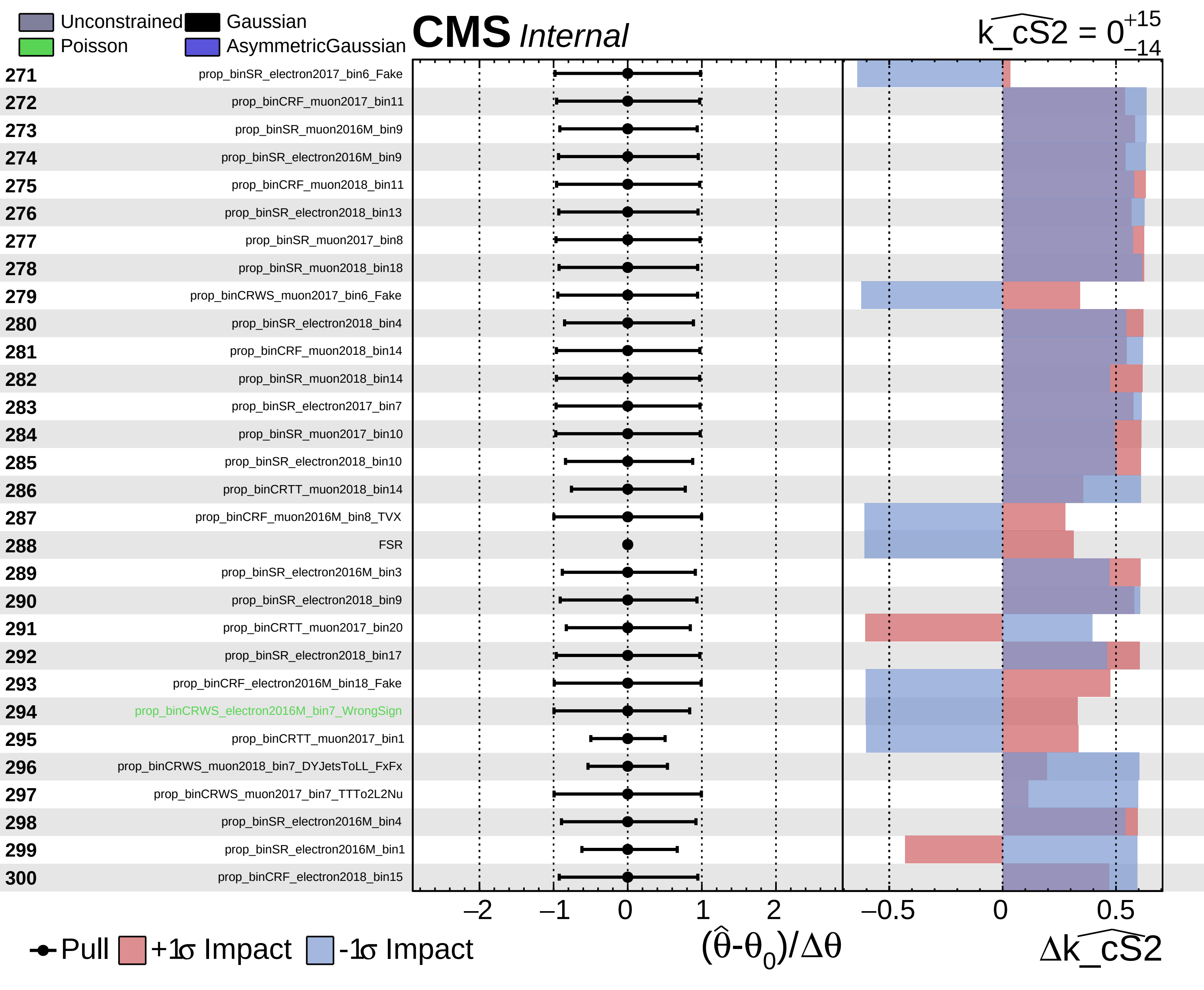
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
181 prop_binCRTT_muon2016M_bin4
182 prop_binCRF_muon2016M_bin13_sm
183 prop_binCRF_muon2017_bin4
184 prop_binCRTT_electron2016M_bin1
185 prop_binCRWS_electron2018_bin6_Fake
186 prop_binSR_muon2017_bin0
187 prop_binCRWS_electron2016M_bin11_DYJetsToLL_FxFx
188 prop_binSR_electron2016M_bin17
189 prop_binCRTT_electron2018_bin5
190 prop_binCRF_electron2017_bin4
191 prop_binSR_muon2017_bin13
192 prop_binCRTT_muon2018_bin2
193 prop_binCRWS_muon2018_bin4
194 prop_binCRWS_electron2017_bin6
195 prop_binCRTT_electron2017_bin18
196 prop_binCRWS_electron2016M_bin5_Fake
197 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
198 prop_binCRTT_electron2016M_bin0
199 prop_binCRWS_electron2017_bin9_Fake
200 prop_binSR_electron2017_bin4
201 prop_binCRTT_muon2017_bin2
202 prop_binCRWS_muon2017_bin8_DYJetsToLL_FxFx
203 prop_binCRTT_electron2016M_bin3
204 prop_binSR_electron2016M_bin7
205 prop_binCRWS_muon2017_bin8_Fake
206 prop_binCRTT_muon2018_bin7
207 prop_binCRWS_muon2017_bin9
208 prop_binCRF_electron2017_bin16_sm
209 prop_binCRF_muon2017_bin14
210 prop_binCRWS_muon2016M_bin7_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



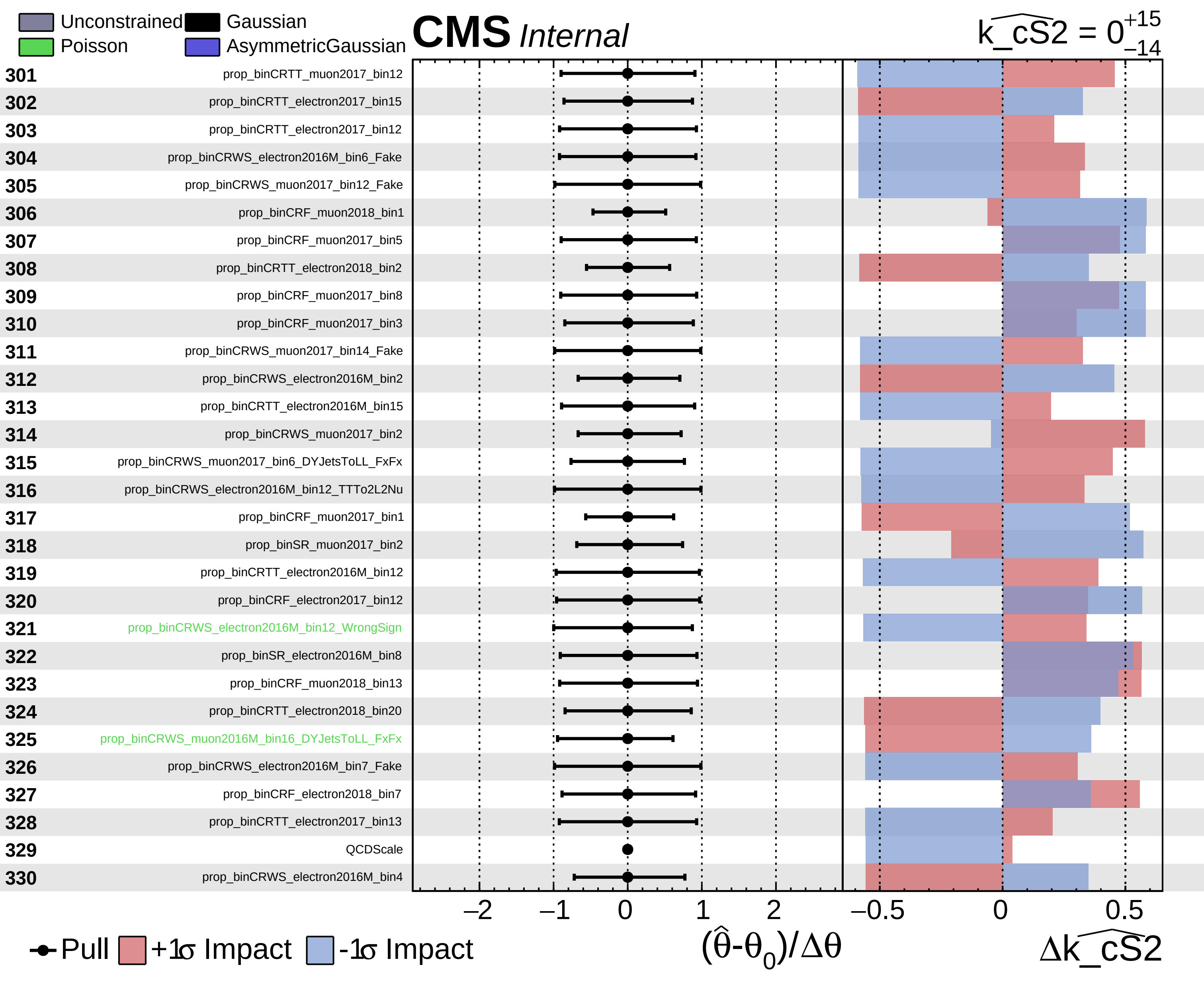
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
211 FR_sys_muon_2016M
212 prop_binSR_muon2017_bin16
213 prop_binCRF_electron2017_bin14
214 prop_binSR_muon2017_bin9
215 prop_binSR_electron2017_bin5
216 prop_binCRWS_muon2016M_bin13_Fake
217 prop_binSR_muon2018_bin19
218 jer_2017
219 prop_binSR_muon2018_bin16
220 prop_binCRTT_electron2016M_bin11
221 prop_binSR_muon2018_bin17
222 prop_binSR_electron2017_bin9
223 prop_binSR_electron2016M_bin0
224 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
225 prop_binSR_electron2018_bin1
226 prop_binSR_muon2016M_bin7
227 prop_binCRF_muon2017_bin6
228 FR_sys_muon_2017
229 ISR
230 prop_binSR_electron2016M_bin15
231 lep_2017
232 prop_binSR_muon2018_bin10
233 prop_binCRTT_electron2016M_bin2
234 prop_binCRF_electron2018_bin17
235 prop_binCRF_electron2016M_bin3
236 prop_binSR_muon2016M_bin10
237 prop_binCRWS_muon2016M_bin18
238 prop_binCRF_muon2018_bin8
239 prop_binSR_muon2016M_bin17
240 prop_binCRTT_muon2016M_bin12
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



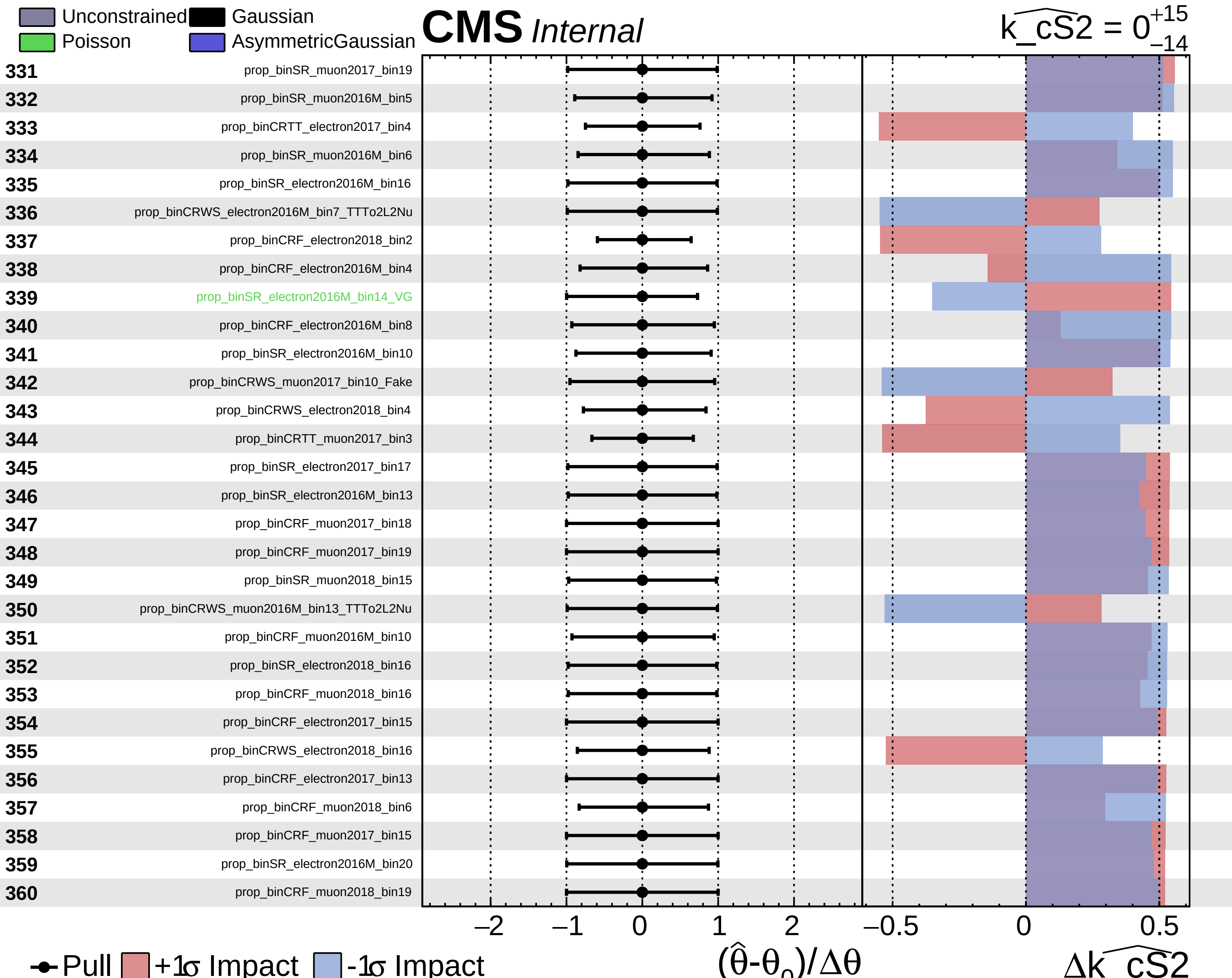
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
241 prop_binSR_muon2017_bin6
242 prop_binSR_electron2018_bin12
243 prop_binSR_electron2018_bin7
244 prop_binSR_muon2017_bin7
245 prop_binSR_electron2017_bin8
246 prop_binSR_muon2017_bin12
247 prop_binSR_muon2018_bin4
248 prop_binCRTT_electron2017_bin3
249 prop_binSR_electron2016M_bin6
250 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
251 prop_binSR_muon2018_bin12
252 prop_binSR_muon2018_bin13
253 prop_binCRWS_muon2016M_bin4
254 prop_binCRTT_electron2018_bin14
255 prop_binCRF_muon2017_bin7
256 prop_binSR_muon2018_bin7
257 prop_binCRTT_electron2018_bin4
258 prop_binCRF_electron2016M_bin2
259 prop_binSR_muon2018_bin6
260 prop_binSR_muon2016M_bin8
261 prop_binSR_electron2018_bin8
262 prop_binSR_electron2018_bin6
263 prop_binSR_electron2018_bin19
264 prop_binSR_electron2018_bin11
265 prop_binSR_muon2018_bin9
266 prop_binSR_muon2017_bin3
267 prop_binSR_electron2016M_bin19
268 prop_binSR_electron2017_bin13
269 tau_vsjet_2016M
270 prop_binCRTT_electron2018_bin19
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



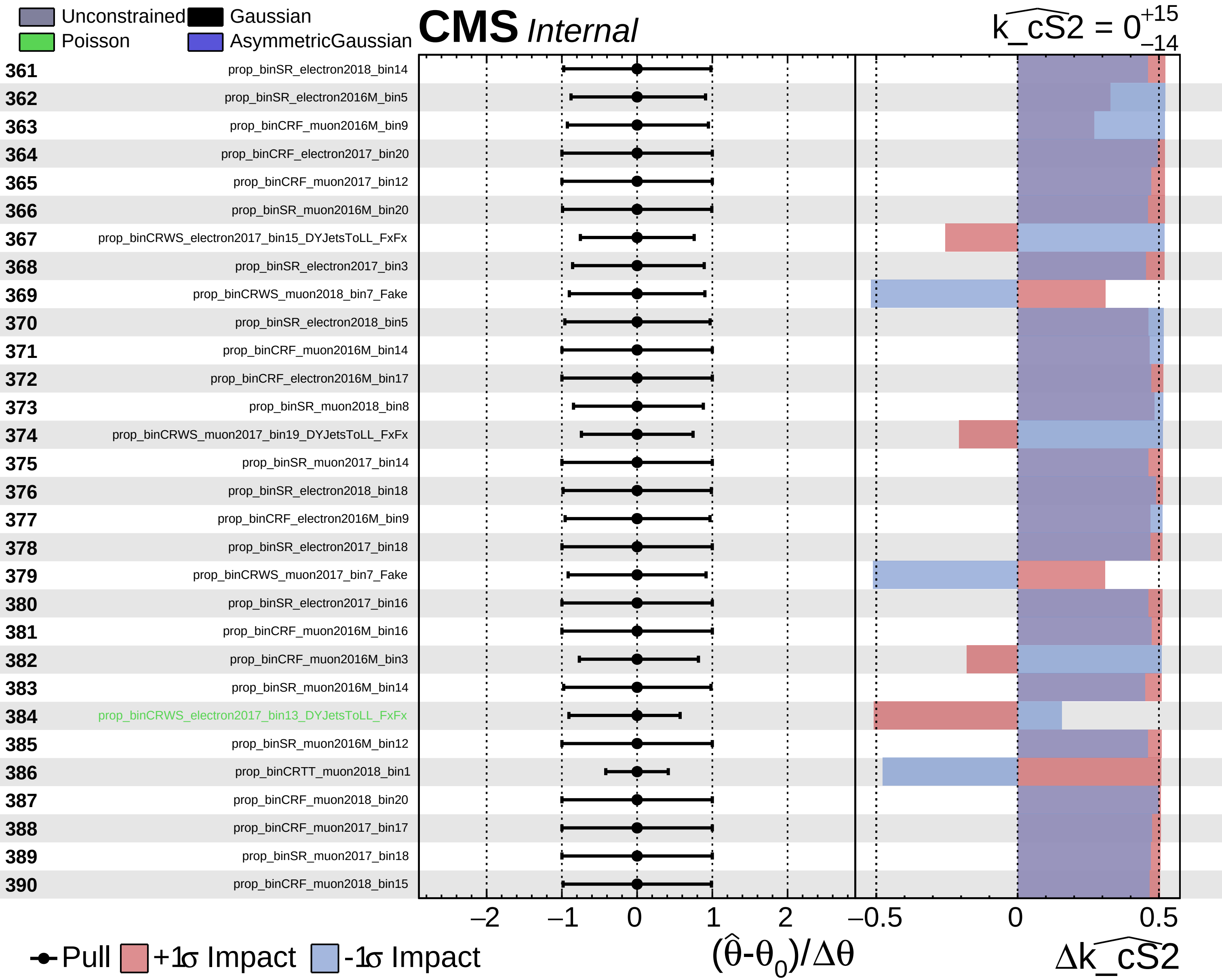
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
271 prop_binSR_electron2017_bin6_Fake
272 prop_binCRF_muon2017_bin11
273 prop_binSR_muon2016M_bin9
274 prop_binSR_electron2016M_bin9
275 prop_binCRF_muon2018_bin11
276 prop_binSR_electron2018_bin13
277 prop_binSR_muon2017_bin8
278 prop_binSR_muon2018_bin18
279 prop_binCRWS_muon2017_bin6_Fake
280 prop_binSR_electron2018_bin4
281 prop_binCRF_muon2018_bin14
282 prop_binSR_muon2018_bin14
283 prop_binSR_electron2017_bin7
284 prop_binSR_muon2017_bin10
285 prop_binSR_electron2018_bin10
286 prop_binCRTT_muon2018_bin14
287 prop_binCRF_muon2016M_bin8_TVX
288 FSR
289 prop_binSR_electron2016M_bin3
290 prop_binSR_electron2018_bin9
291 prop_binCRTT_muon2017_bin20
292 prop_binSR_electron2018_bin17
293 prop_binCRF_electron2016M_bin18_Fake
294 prop_binCRWS_electron2016M_bin7_WrongSign
295 prop_binCRTT_muon2017_bin1
296 prop_binCRWS_muon2018_bin7_DYJetsToLL_FxFx
297 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
298 prop_binSR_electron2016M_bin4
299 prop_binSR_electron2016M_bin1
300 prop_binCRF_electron2018_bin15
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

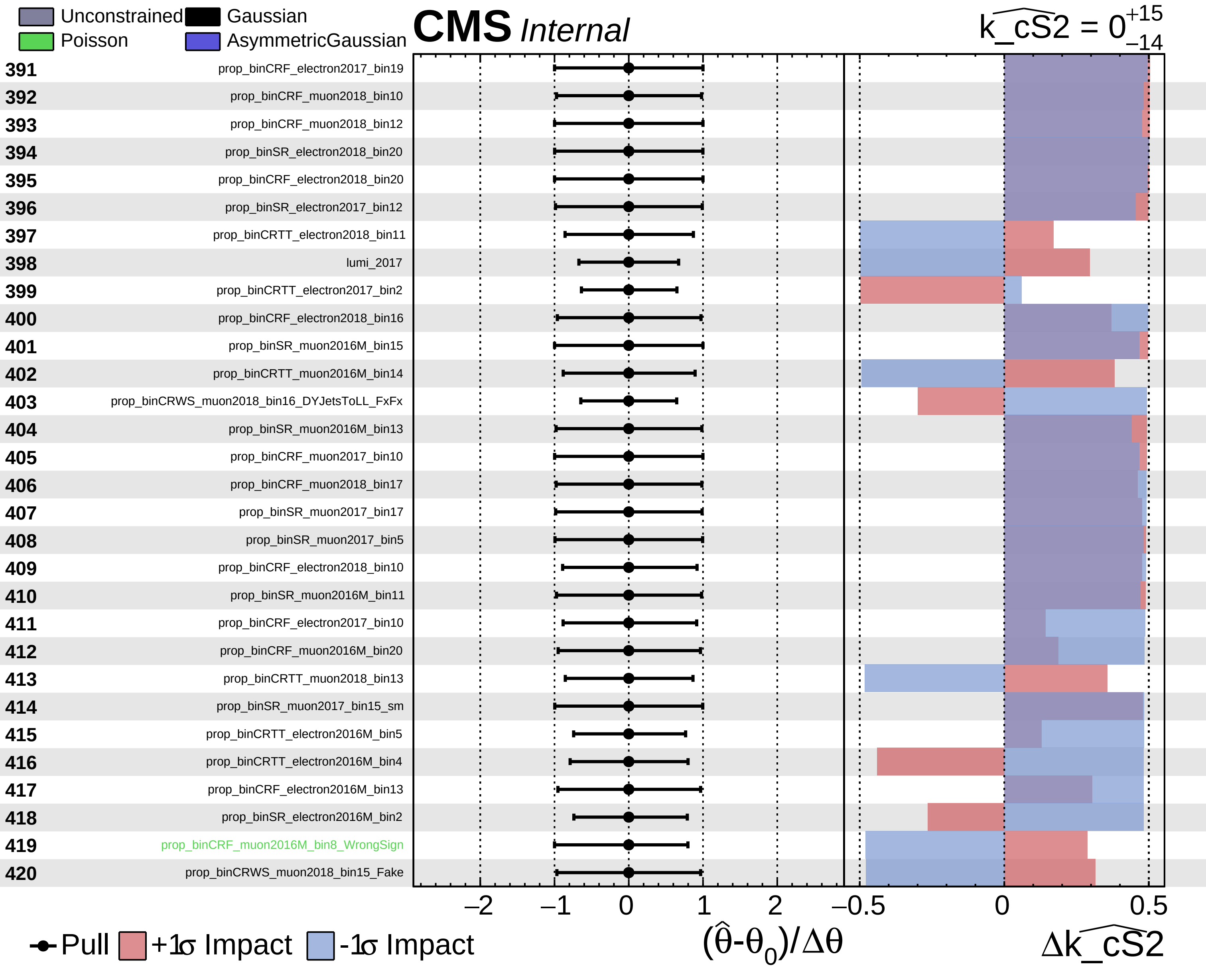


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
301 prop_binCRTT_muon2017_bin12
302 prop_binCRTT_electron2017_bin15
303 prop_binCRTT_electron2017_bin12
304 prop_binCRWS_electron2016M_bin6_Fake
305 prop_binCRWS_muon2017_bin12_Fake
306 prop_binCRF_muon2018_bin1
307 prop_binCRF_muon2017_bin5
308 prop_binCRTT_electron2018_bin2
309 prop_binCRF_muon2017_bin8
310 prop_binCRF_muon2017_bin3
311 prop_binCRWS_muon2017_bin14_Fake
312 prop_binCRWS_electron2016M_bin2
313 prop_binCRTT_electron2016M_bin15
314 prop_binCRWS_muon2017_bin2
315 prop_binCRWS_muon2017_bin6_DYJetsToLL_FxFx
316 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
317 prop_binCRF_muon2017_bin1
318 prop_binSR_muon2017_bin2
319 prop_binCRTT_electron2016M_bin12
320 prop_binCRF_electron2017_bin12
321 prop_binCRWS_electron2016M_bin12_WrongSign
322 prop_binSR_electron2016M_bin8
323 prop_binCRF_muon2018_bin13
324 prop_binCRTT_electron2018_bin20
325 prop_binCRWS_muon2016M_bin16_DYJetsToLL_FxFx
326 prop_binCRWS_electron2016M_bin7_Fake
327 prop_binCRF_electron2018_bin7
328 prop_binCRTT_electron2017_bin13
329 QCDScale
330 prop_binCRWS_electron2016M_bin4
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cS2

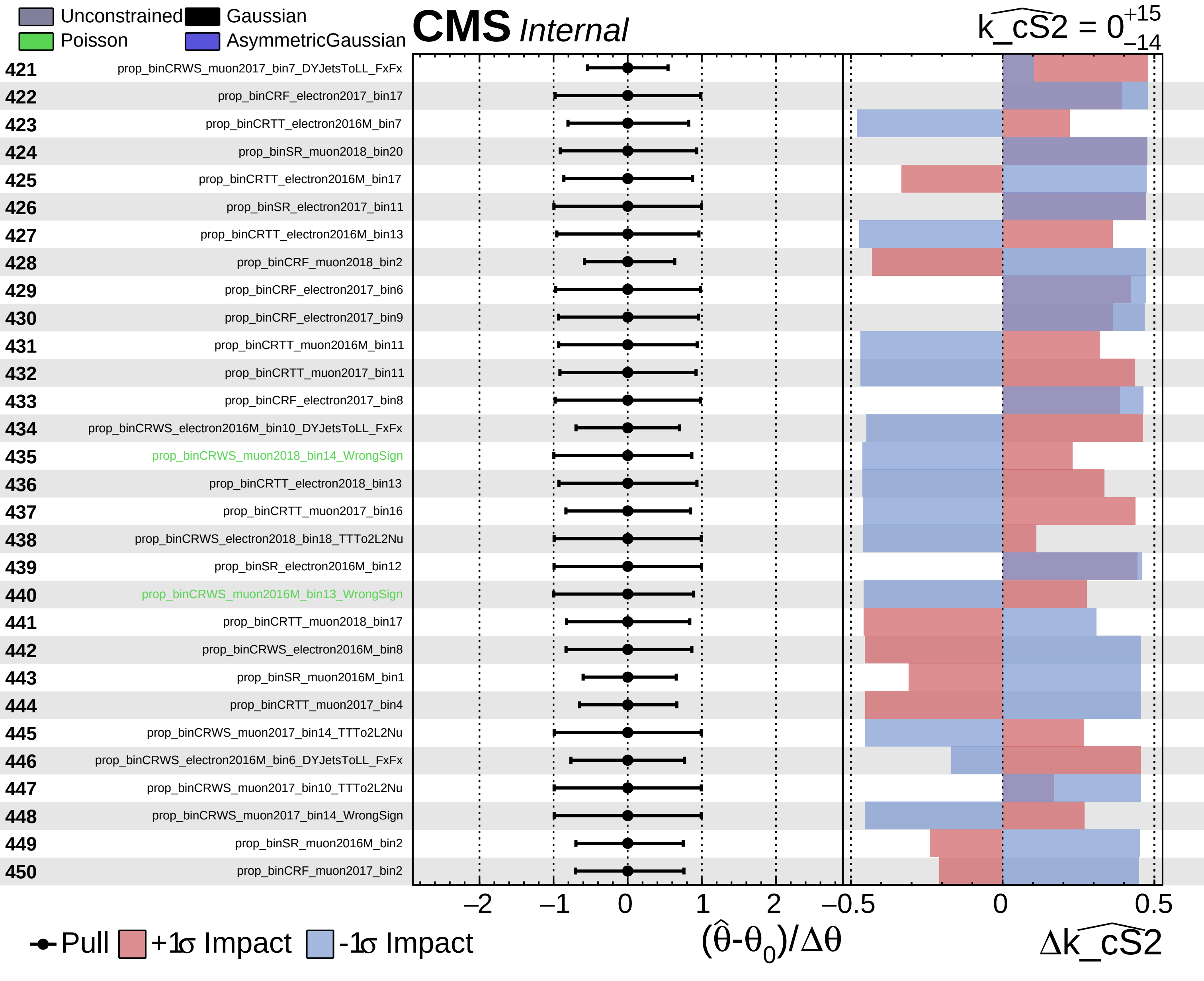


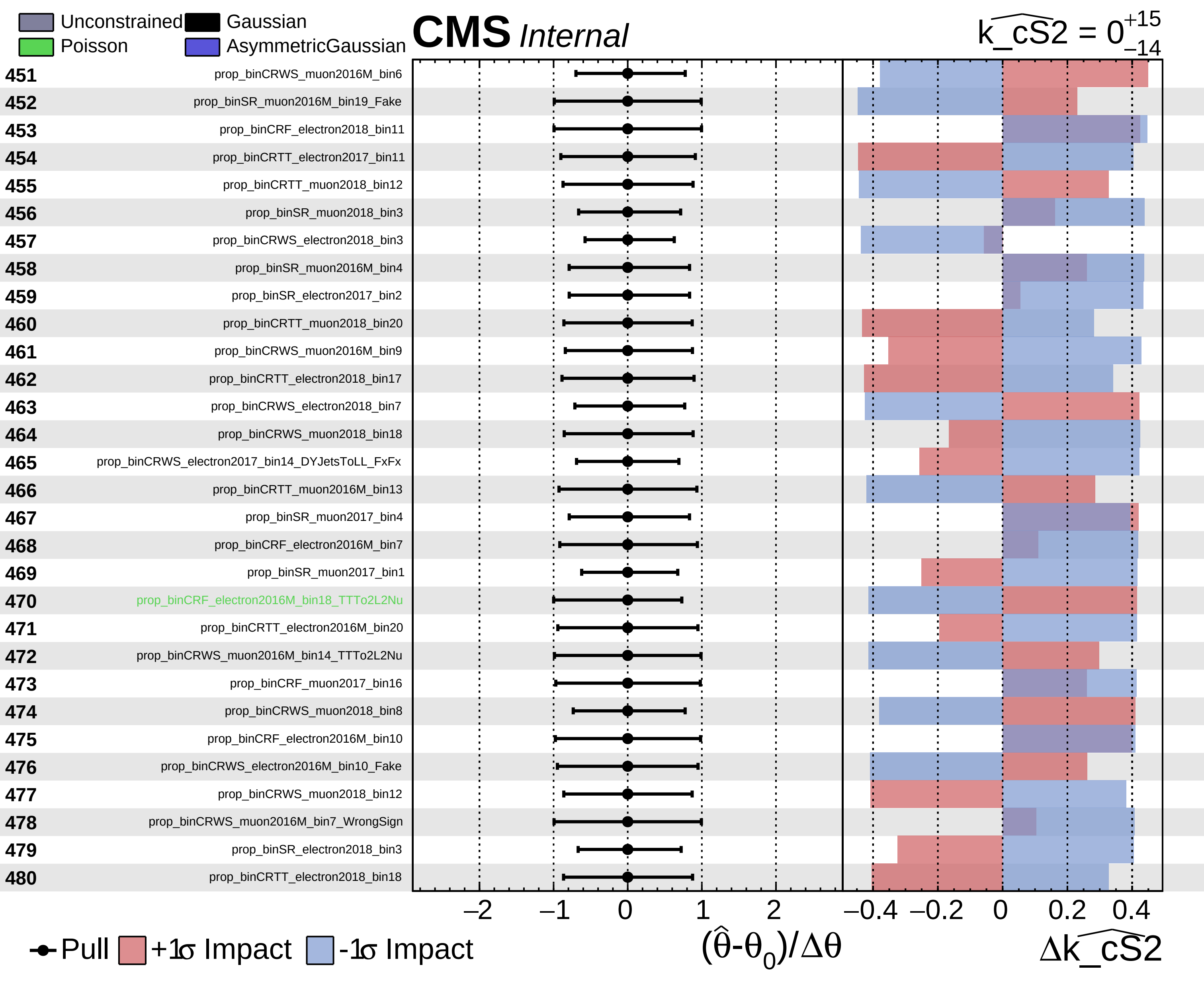
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
331 prop_binSR_muon2017_bin19
332 prop_binSR_muon2016M_bin5
333 prop_binCRTT_electron2017_bin4
334 prop_binSR_muon2016M_bin6
335 prop_binSR_electron2016M_bin16
336 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
337 prop_binCRF_electron2018_bin2
338 prop_binCRF_electron2016M_bin4
339 prop_binSR_electron2016M_bin14_VG
340 prop_binCRF_electron2016M_bin8
341 prop_binSR_electron2016M_bin10
342 prop_binCRWS_muon2017_bin10_Fake
343 prop_binCRWS_electron2018_bin4
344 prop_binCRTT_muon2017_bin3
345 prop_binSR_electron2017_bin17
346 prop_binSR_electron2016M_bin13
347 prop_binCRF_muon2017_bin18
348 prop_binCRF_muon2017_bin19
349 prop_binSR_muon2018_bin15
350 prop_binCRWS_muon2016M_bin13_TTTo2L2Nu
351 prop_binCRF_muon2016M_bin10
352 prop_binSR_electron2018_bin16
353 prop_binCRF_muon2018_bin16
354 prop_binCRF_electron2017_bin15
355 prop_binCRWS_electron2018_bin16
356 prop_binCRF_electron2017_bin13
357 prop_binCRF_muon2018_bin6
358 prop_binCRF_muon2017_bin15
359 prop_binSR_electron2016M_bin20
360 prop_binCRF_muon2018_bin19
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
361 prop_binSR_electron2018_bin14
362 prop_binSR_electron2016M_bin5
363 prop_binCRF_muon2016M_bin9
364 prop_binCRF_electron2017_bin20
365 prop_binCRF_muon2017_bin12
366 prop_binSR_muon2016M_bin20
367 prop_binCRWS_electron2017_bin15_DYJetsToLL_FxFx
368 prop_binSR_electron2017_bin3
369 prop_binCRWS_muon2018_bin7_Fake
370 prop_binSR_electron2018_bin5
371 prop_binCRF_muon2016M_bin14
372 prop_binCRF_electron2016M_bin17
373 prop_binSR_muon2018_bin8
374 prop_binCRWS_muon2017_bin19_DYJetsToLL_FxFx
375 prop_binSR_muon2017_bin14
376 prop_binSR_electron2018_bin18
377 prop_binCRF_electron2016M_bin9
378 prop_binSR_electron2017_bin18
379 prop_binCRWS_muon2017_bin7_Fake
380 prop_binSR_electron2017_bin16
381 prop_binCRF_muon2016M_bin16
382 prop_binCRF_muon2016M_bin3
383 prop_binSR_muon2016M_bin14
384 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
385 prop_binSR_muon2016M_bin12
386 prop_binCRTT_muon2018_bin1
387 prop_binCRF_muon2018_bin20
388 prop_binCRF_muon2017_bin17
389 prop_binSR_muon2017_bin18
390 prop_binCRF_muon2018_bin15
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

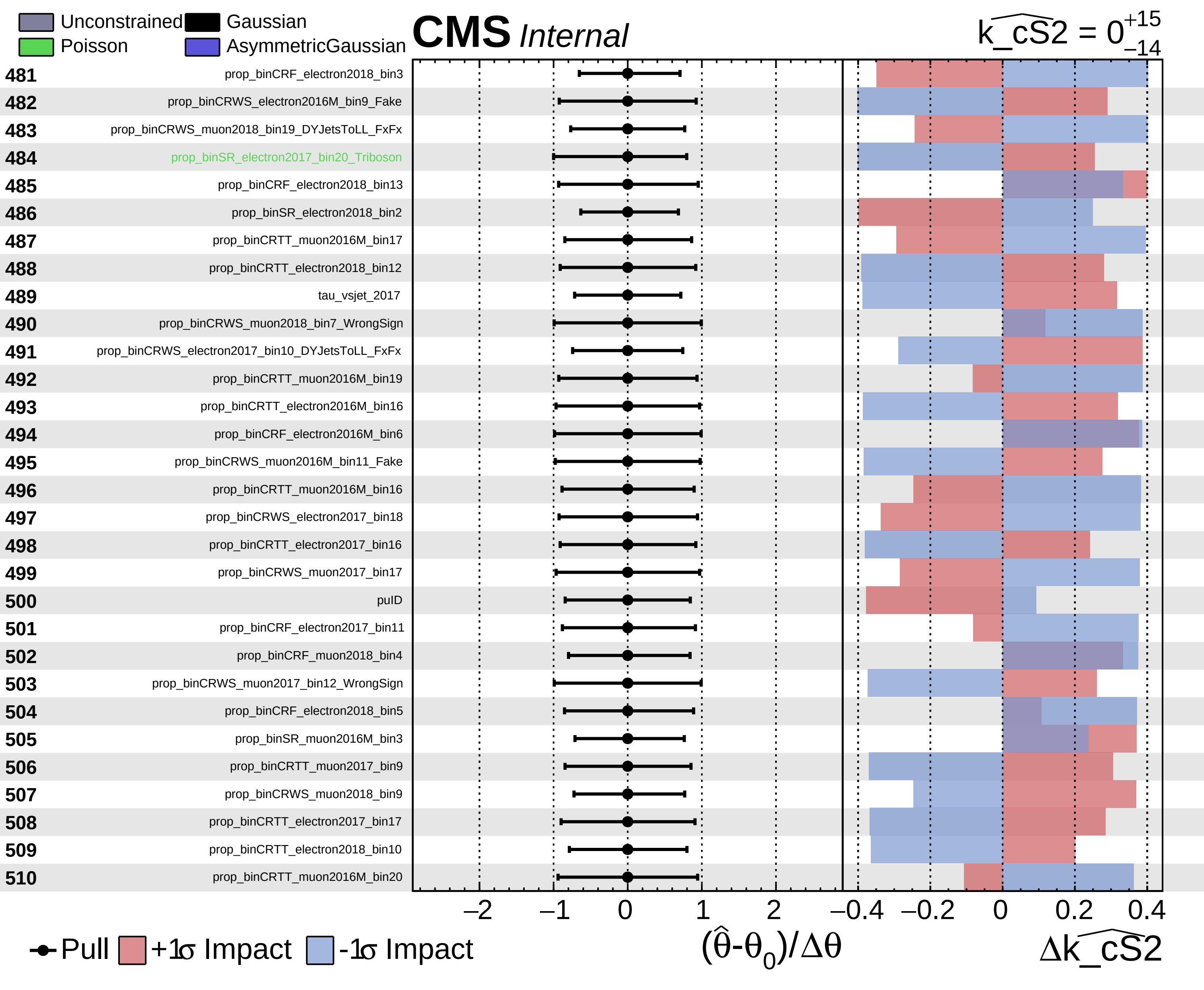


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
391 prop_binCRF_electron2017_bin19
392 prop_binCRF_muon2018_bin10
393 prop_binCRF_muon2018_bin12
394 prop_binSR_electron2018_bin20
395 prop_binCRF_electron2018_bin20
396 prop_binSR_electron2017_bin12
397 prop_binCRTT_electron2018_bin11
398 lumi_2017
399 prop_binCRTT_electron2017_bin2
400 prop_binCRF_electron2018_bin16
401 prop_binSR_muon2016M_bin15
402 prop_binCRTT_muon2016M_bin14
403 prop_binCRWS_muon2018_bin16_DYJetsToLL_FxFx
404 prop_binSR_muon2016M_bin13
405 prop_binCRF_muon2017_bin10
406 prop_binCRF_muon2018_bin17
407 prop_binSR_muon2017_bin17
408 prop_binSR_muon2017_bin5
409 prop_binCRF_electron2018_bin10
410 prop_binSR_muon2016M_bin11
411 prop_binCRF_electron2017_bin10
412 prop_binCRF_muon2016M_bin20
413 prop_binCRTT_muon2018_bin13
414 prop_binSR_muon2017_bin15_sm
415 prop_binCRTT_electron2016M_bin5
416 prop_binCRTT_electron2016M_bin4
417 prop_binCRF_electron2016M_bin13
418 prop_binSR_electron2016M_bin2
419 prop_binCRF_muon2016M_bin8_WrongSign
420 prop_binCRWS_muon2018_bin15_Fake
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

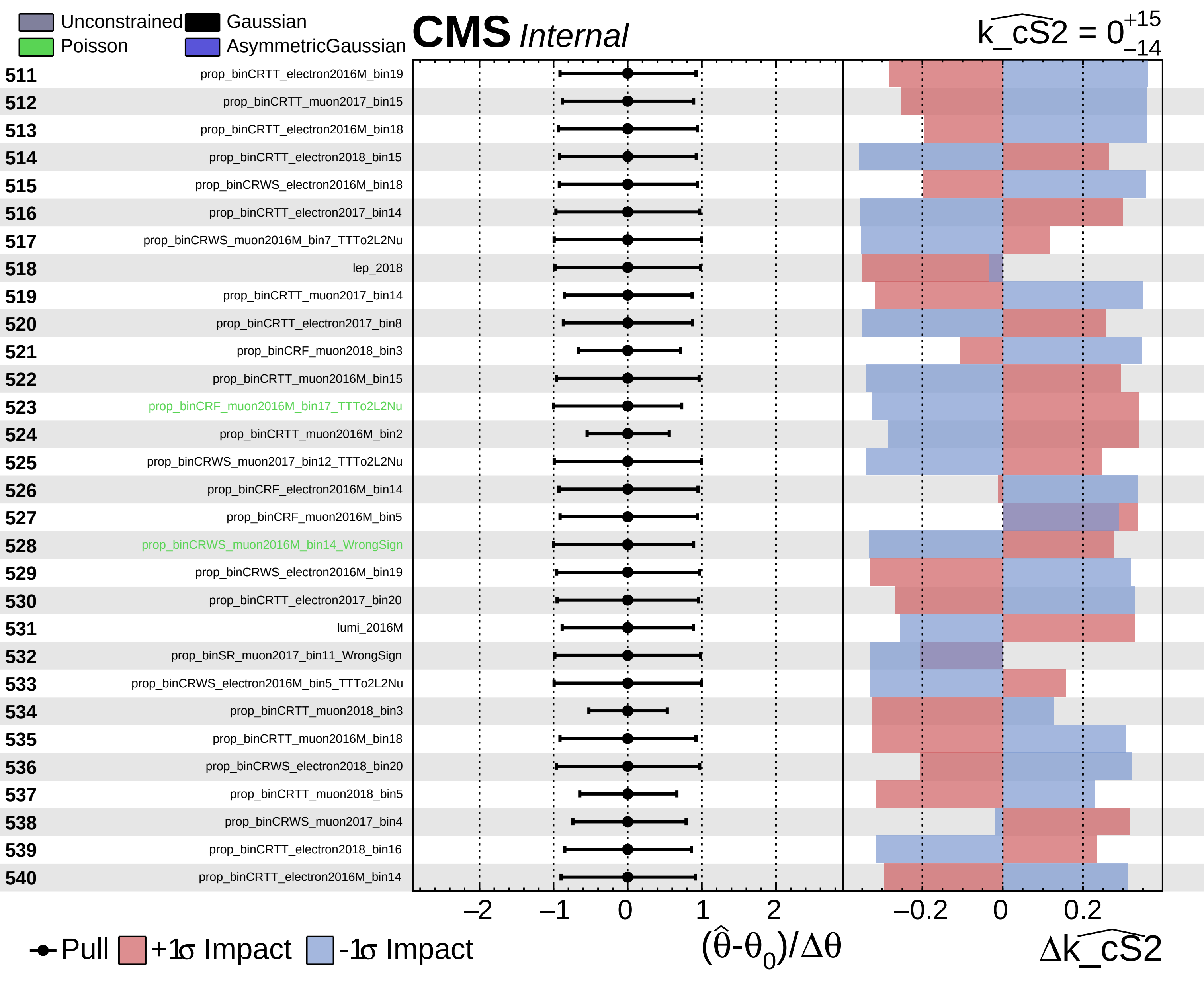

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
421 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
422 prop_binCRF_electron2017_bin17
423 prop_binCRTT_electron2016M_bin7
424 prop_binSR_muon2018_bin20
425 prop_binCRTT_electron2016M_bin17
426 prop_binSR_electron2017_bin11
427 prop_binCRTT_electron2016M_bin13
428 prop_binCRF_muon2018_bin2
429 prop_binCRF_electron2017_bin6
430 prop_binCRF_electron2017_bin9
431 prop_binCRTT_muon2016M_bin11
432 prop_binCRTT_muon2017_bin11
433 prop_binCRF_electron2017_bin8
434 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
435 prop_binCRWS_muon2018_bin14_WrongSign
436 prop_binCRTT_electron2018_bin13
437 prop_binCRTT_muon2017_bin16
438 prop_binCRWS_electron2018_bin18_TTTo2L2Nu
439 prop_binSR_electron2016M_bin12
440 prop_binCRWS_muon2016M_bin13_WrongSign
441 prop_binCRTT_muon2018_bin17
442 prop_binCRWS_electron2016M_bin8
443 prop_binSR_muon2016M_bin1
444 prop_binCRTT_muon2017_bin4
445 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
446 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
447 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
448 prop_binCRWS_muon2017_bin14_WrongSign
449 prop_binSR_muon2016M_bin2
450 prop_binCRF_muon2017_bin2
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



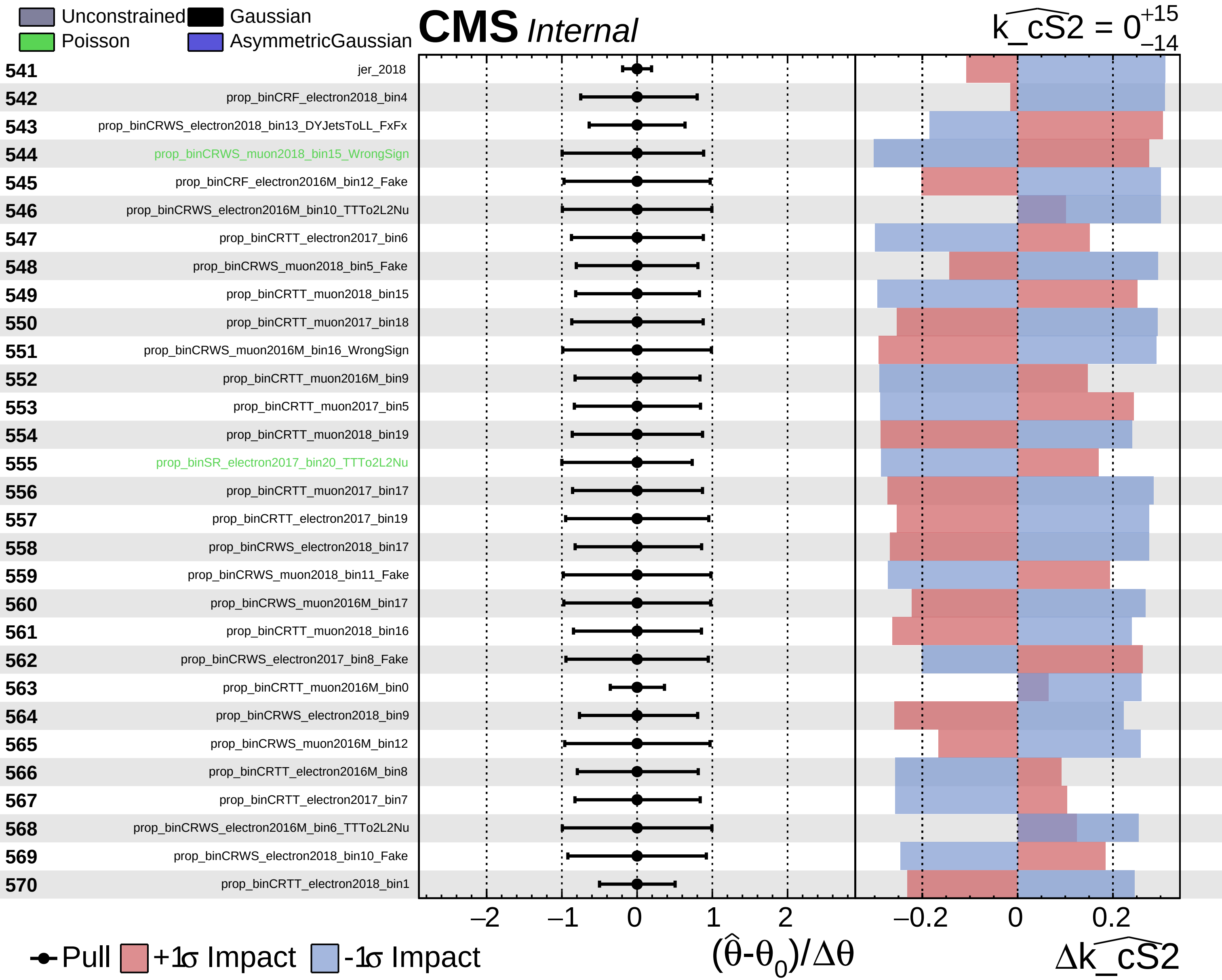
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
451 prop_binCRWS_muon2016M_bin6
452 prop_binSR_muon2016M_bin19_Fake
453 prop_binCRF_electron2018_bin11
454 prop_binCRTT_electron2017_bin11
455 prop_binCRTT_muon2018_bin12
456 prop_binSR_muon2018_bin3
457 prop_binCRWS_electron2018_bin3
458 prop_binSR_muon2016M_bin4
459 prop_binSR_electron2017_bin2
460 prop_binCRTT_muon2018_bin20
461 prop_binCRWS_muon2016M_bin9
462 prop_binCRTT_electron2018_bin17
463 prop_binCRWS_electron2018_bin7
464 prop_binCRWS_muon2018_bin18
465 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
466 prop_binCRTT_muon2016M_bin13
467 prop_binSR_muon2017_bin4
468 prop_binCRF_electron2016M_bin7
469 prop_binSR_muon2017_bin1
470 prop_binCRF_electron2016M_bin18_TTTo2L2Nu
471 prop_binCRTT_electron2016M_bin20
472 prop_binCRWS_muon2016M_bin14_TTTo2L2Nu
473 prop_binCRF_muon2017_bin16
474 prop_binCRWS_muon2018_bin8
475 prop_binCRF_electron2016M_bin10
476 prop_binCRWS_electron2016M_bin10_Fake
477 prop_binCRWS_muon2018_bin12
478 prop_binCRWS_muon2016M_bin7_WrongSign
479 prop_binSR_electron2018_bin3
480 prop_binCRTT_electron2018_bin18
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

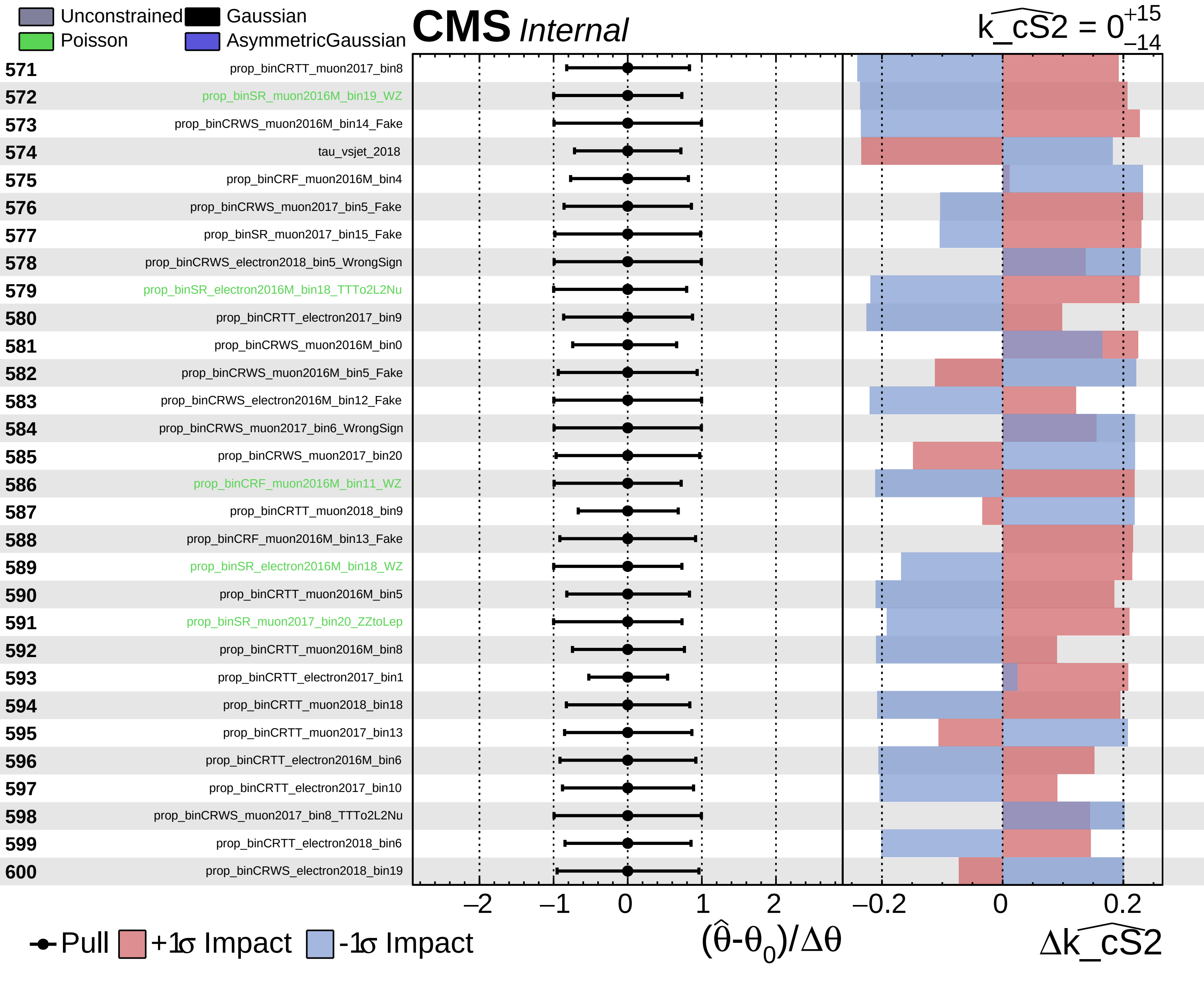


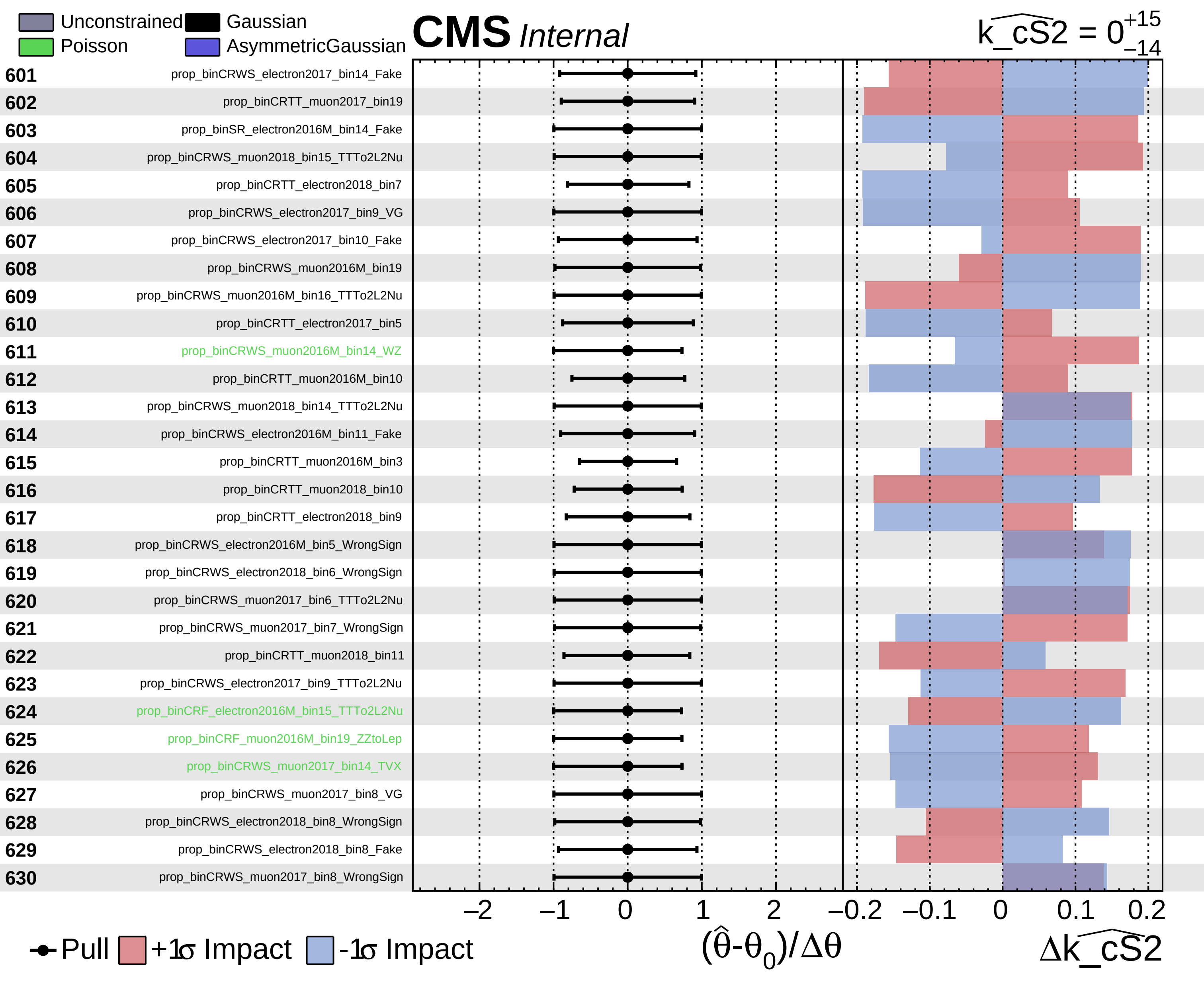
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
481 prop_binCRF_electron2018_bin3
482 prop_binCRWS_electron2016M_bin9_Fake
483 prop_binCRWS_muon2018_bin19_DYJetsToLL_FxFx
484 prop_binSR_electron2017_bin20_Triboson
485 prop_binCRF_electron2018_bin13
486 prop_binSR_electron2018_bin2
487 prop_binCRTT_muon2016M_bin17
488 prop_binCRTT_electron2018_bin12
489 tau_vsjet_2017
490 prop_binCRWS_muon2018_bin7_WrongSign
491 prop_binCRWS_electron2017_bin10_DYJetsToLL_FxFx
492 prop_binCRTT_muon2016M_bin19
493 prop_binCRTT_electron2016M_bin16
494 prop_binCRF_electron2016M_bin6
495 prop_binCRWS_muon2016M_bin11_Fake
496 prop_binCRTT_muon2016M_bin16
497 prop_binCRWS_electron2017_bin18
498 prop_binCRTT_electron2017_bin16
499 prop_binCRWS_muon2017_bin17
500 puID
501 prop_binCRF_electron2017_bin11
502 prop_binCRF_muon2018_bin4
503 prop_binCRWS_muon2017_bin12_WrongSign
504 prop_binCRF_electron2018_bin5
505 prop_binSR_muon2016M_bin3
506 prop_binCRTT_muon2017_bin9
507 prop_binCRWS_muon2018_bin9
508 prop_binCRTT_electron2017_bin17
509 prop_binCRTT_electron2018_bin10
510 prop_binCRTT_muon2016M_bin20
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



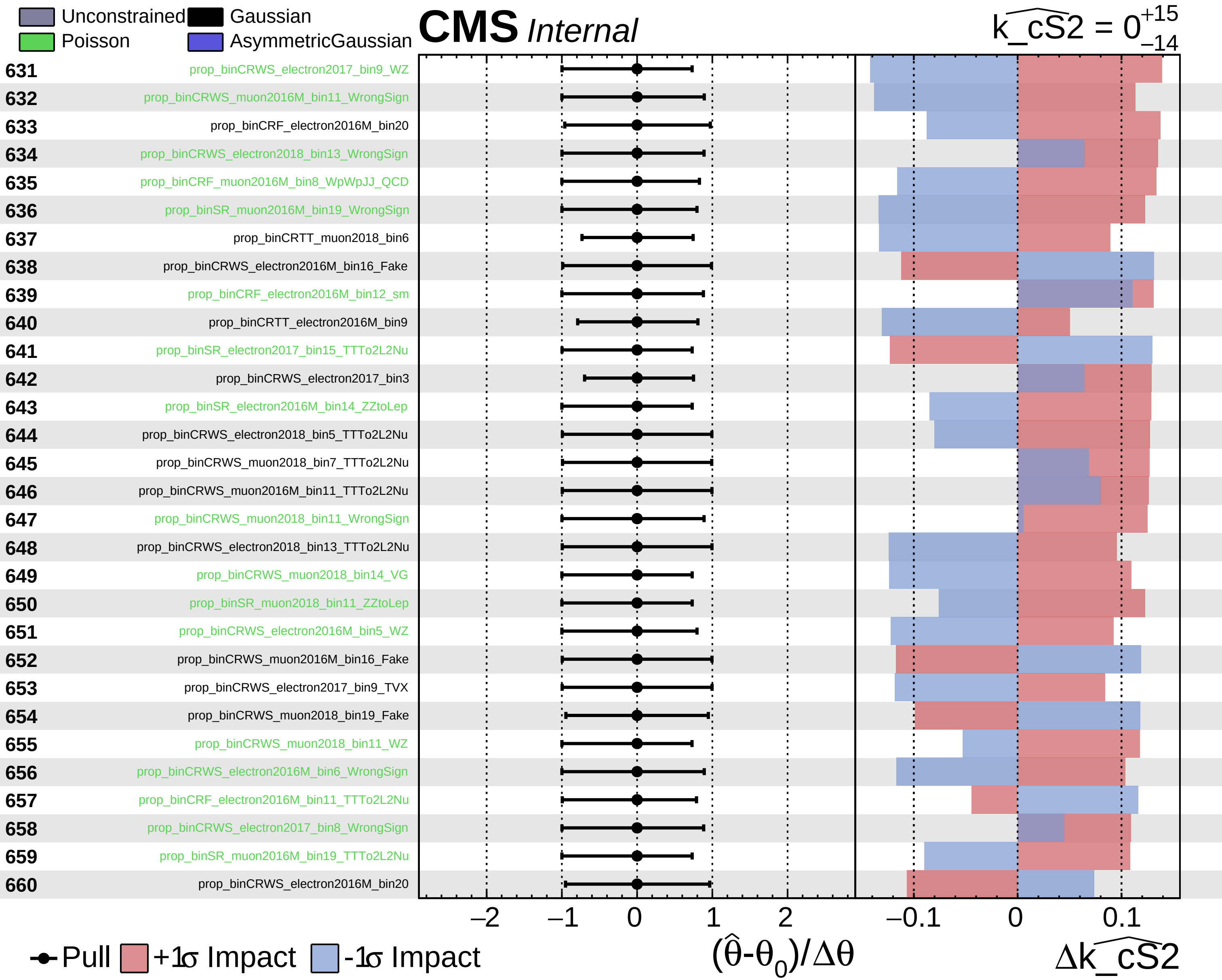
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
511 prop_binCRTT_electron2016M_bin19
512 prop_binCRTT_muon2017_bin15
513 prop_binCRTT_electron2016M_bin18
514 prop_binCRTT_electron2018_bin15
515 prop_binCRWS_electron2016M_bin18
516 prop_binCRTT_electron2017_bin14
517 prop_binCRWS_muon2016M_bin7_TTTo2L2Nu
518 lep_2018
519 prop_binCRTT_muon2017_bin14
520 prop_binCRTT_electron2017_bin8
521 prop_binCRF_muon2018_bin3
522 prop_binCRTT_muon2016M_bin15
523 prop_binCRF_muon2016M_bin17_TTTo2L2Nu
524 prop_binCRTT_muon2016M_bin2
525 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
526 prop_binCRF_electron2016M_bin14
527 prop_binCRF_muon2016M_bin5
528 prop_binCRWS_muon2016M_bin14_WrongSign
529 prop_binCRWS_electron2016M_bin19
530 prop_binCRTT_electron2017_bin20
531 lumi_2016M
532 prop_binSR_muon2017_bin11_WrongSign
533 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
534 prop_binCRTT_muon2018_bin3
535 prop_binCRTT_muon2016M_bin18
536 prop_binCRWS_electron2018_bin20
537 prop_binCRTT_muon2018_bin5
538 prop_binCRWS_muon2017_bin4
539 prop_binCRTT_electron2018_bin16
540 prop_binCRTT_electron2016M_bin14
−2
−1
0
1
2
−0.2
0
0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
541 jer_2018
542 prop_binCRF_electron2018_bin4
543 prop_binCRWS_electron2018_bin13_DYJetsToLL_FxFx
544 prop_binCRWS_muon2018_bin15_WrongSign
545 prop_binCRF_electron2016M_bin12_Fake
546 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
547 prop_binCRTT_electron2017_bin6
548 prop_binCRWS_muon2018_bin5_Fake
549 prop_binCRTT_muon2018_bin15
550 prop_binCRTT_muon2017_bin18
551 prop_binCRWS_muon2016M_bin16_WrongSign
552 prop_binCRTT_muon2016M_bin9
553 prop_binCRTT_muon2017_bin5
554 prop_binCRTT_muon2018_bin19
555 prop_binSR_electron2017_bin20_TTTo2L2Nu
556 prop_binCRTT_muon2017_bin17
557 prop_binCRTT_electron2017_bin19
558 prop_binCRWS_electron2018_bin17
559 prop_binCRWS_muon2018_bin11_Fake
560 prop_binCRWS_muon2016M_bin17
561 prop_binCRTT_muon2018_bin16
562 prop_binCRWS_electron2017_bin8_Fake
563 prop_binCRTT_muon2016M_bin0
564 prop_binCRWS_electron2018_bin9
565 prop_binCRWS_muon2016M_bin12
566 prop_binCRTT_electron2016M_bin8
567 prop_binCRTT_electron2017_bin7
568 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
569 prop_binCRWS_electron2018_bin10_Fake
570 prop_binCRTT_electron2018_bin1
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

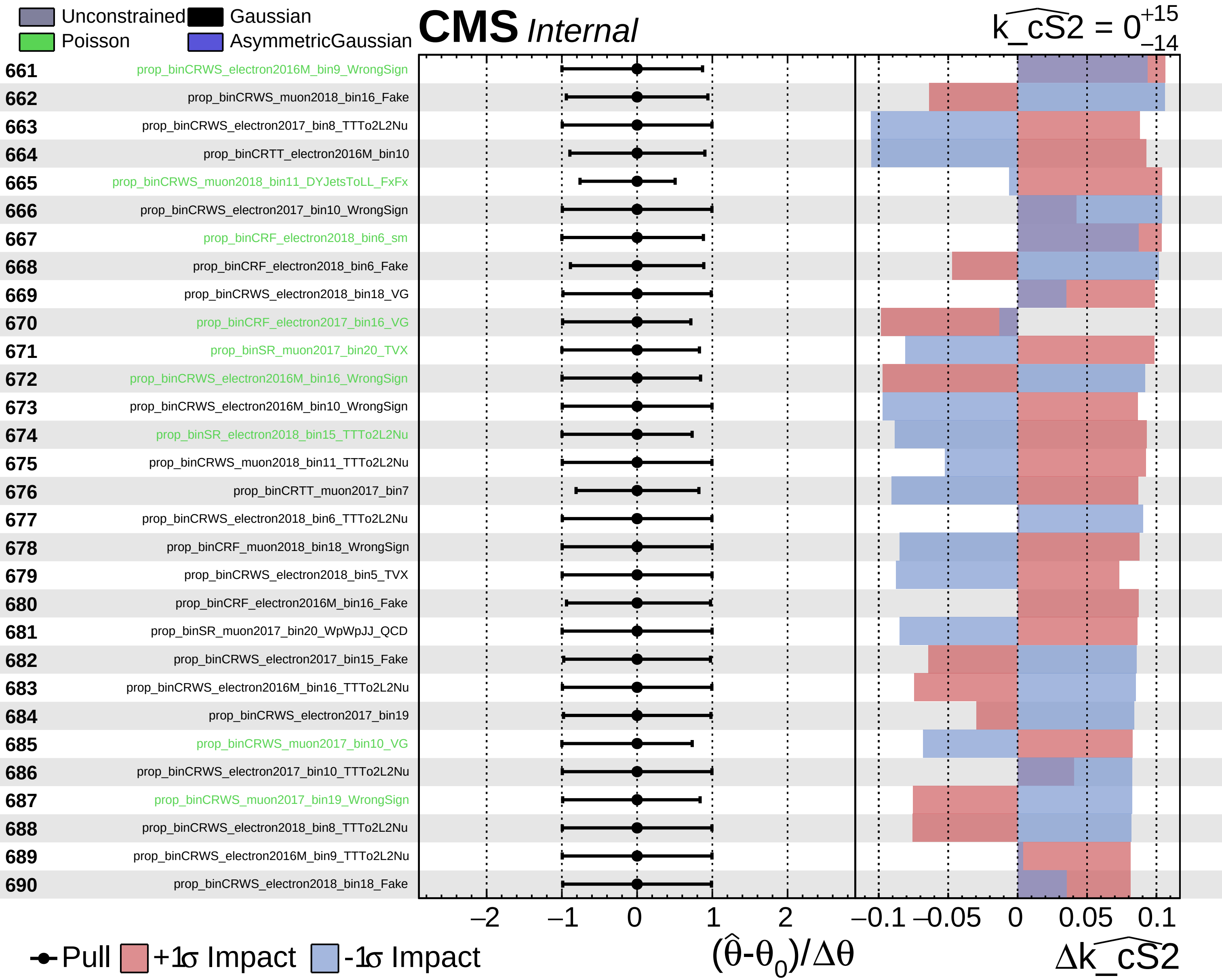

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
571 prop_binCRTT_muon2017_bin8
572 prop_binSR_muon2016M_bin19_WZ
573 prop_binCRWS_muon2016M_bin14_Fake
574 tau_vsjet_2018
575 prop_binCRF_muon2016M_bin4
576 prop_binCRWS_muon2017_bin5_Fake
577 prop_binSR_muon2017_bin15_Fake
578 prop_binCRWS_electron2018_bin5_WrongSign
579 prop_binSR_electron2016M_bin18_TTTo2L2Nu
580 prop_binCRTT_electron2017_bin9
581 prop_binCRWS_muon2016M_bin0
582 prop_binCRWS_muon2016M_bin5_Fake
583 prop_binCRWS_electron2016M_bin12_Fake
584 prop_binCRWS_muon2017_bin6_WrongSign
585 prop_binCRWS_muon2017_bin20
586 prop_binCRF_muon2016M_bin11_WZ
587 prop_binCRTT_muon2018_bin9
588 prop_binCRF_muon2016M_bin13_Fake
589 prop_binSR_electron2016M_bin18_WZ
590 prop_binCRTT_muon2016M_bin5
591 prop_binSR_muon2017_bin20_ZZtoLep
592 prop_binCRTT_muon2016M_bin8
593 prop_binCRTT_electron2017_bin1
594 prop_binCRTT_muon2018_bin18
595 prop_binCRTT_muon2017_bin13
596 prop_binCRTT_electron2016M_bin6
597 prop_binCRTT_electron2017_bin10
598 prop_binCRWS_muon2017_bin8_TTTo2L2Nu
599 prop_binCRTT_electron2018_bin6
600 prop_binCRWS_electron2018_bin19
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2



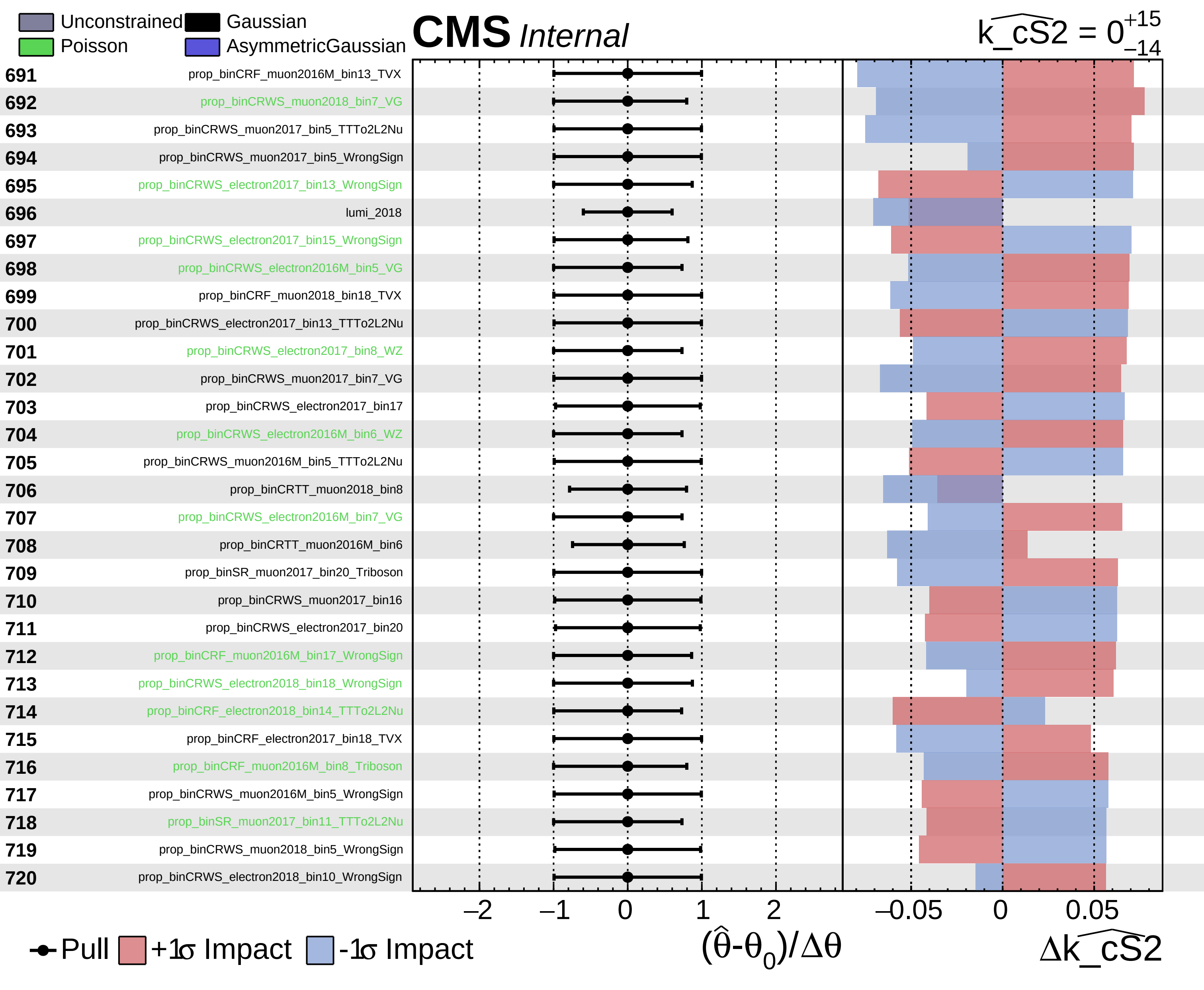
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
601 prop_binCRWS_electron2017_bin14_Fake
602 prop_binCRTT_muon2017_bin19
603 prop_binSR_electron2016M_bin14_Fake
604 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
605 prop_binCRTT_electron2018_bin7
606 prop_binCRWS_electron2017_bin9_VG
607 prop_binCRWS_electron2017_bin10_Fake
608 prop_binCRWS_muon2016M_bin19
609 prop_binCRWS_muon2016M_bin16_TTTo2L2Nu
610 prop_binCRTT_electron2017_bin5
611 prop_binCRWS_muon2016M_bin14_WZ
612 prop_binCRTT_muon2016M_bin10
613 prop_binCRWS_muon2018_bin14_TTTo2L2Nu
614 prop_binCRWS_electron2016M_bin11_Fake
615 prop_binCRTT_muon2016M_bin3
616 prop_binCRTT_muon2018_bin10
617 prop_binCRTT_electron2018_bin9
618 prop_binCRWS_electron2016M_bin5_WrongSign
619 prop_binCRWS_electron2018_bin6_WrongSign
620 prop_binCRWS_muon2017_bin6_TTTo2L2Nu
621 prop_binCRWS_muon2017_bin7_WrongSign
622 prop_binCRTT_muon2018_bin11
623 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
624 prop_binCRF_electron2016M_bin15_TTTo2L2Nu
625 prop_binCRF_muon2016M_bin19_ZZtoLep
626 prop_binCRWS_muon2017_bin14_TVX
627 prop_binCRWS_muon2017_bin8_VG
628 prop_binCRWS_electron2018_bin8_WrongSign
629 prop_binCRWS_electron2018_bin8_Fake
630 prop_binCRWS_muon2017_bin8_WrongSign
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

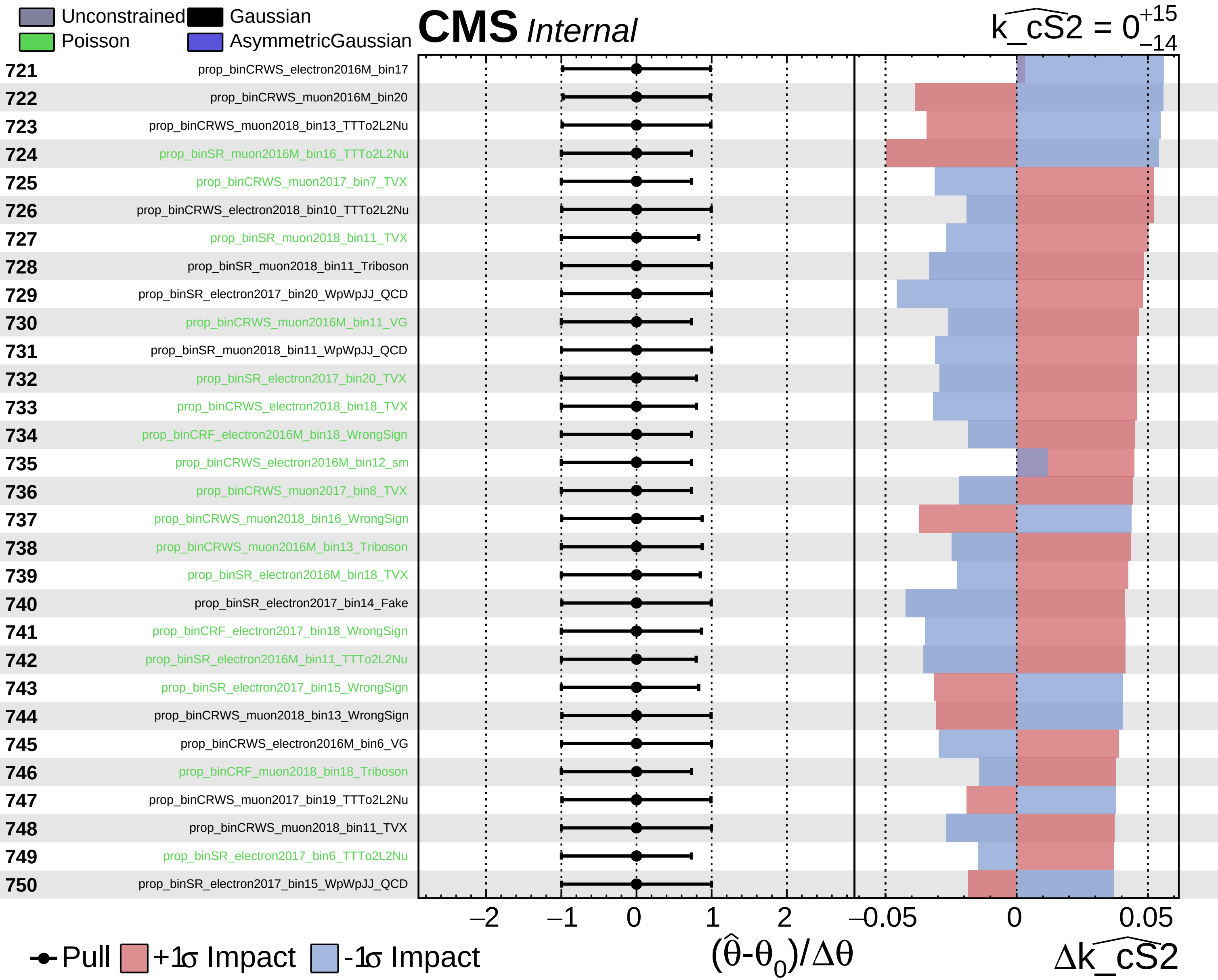


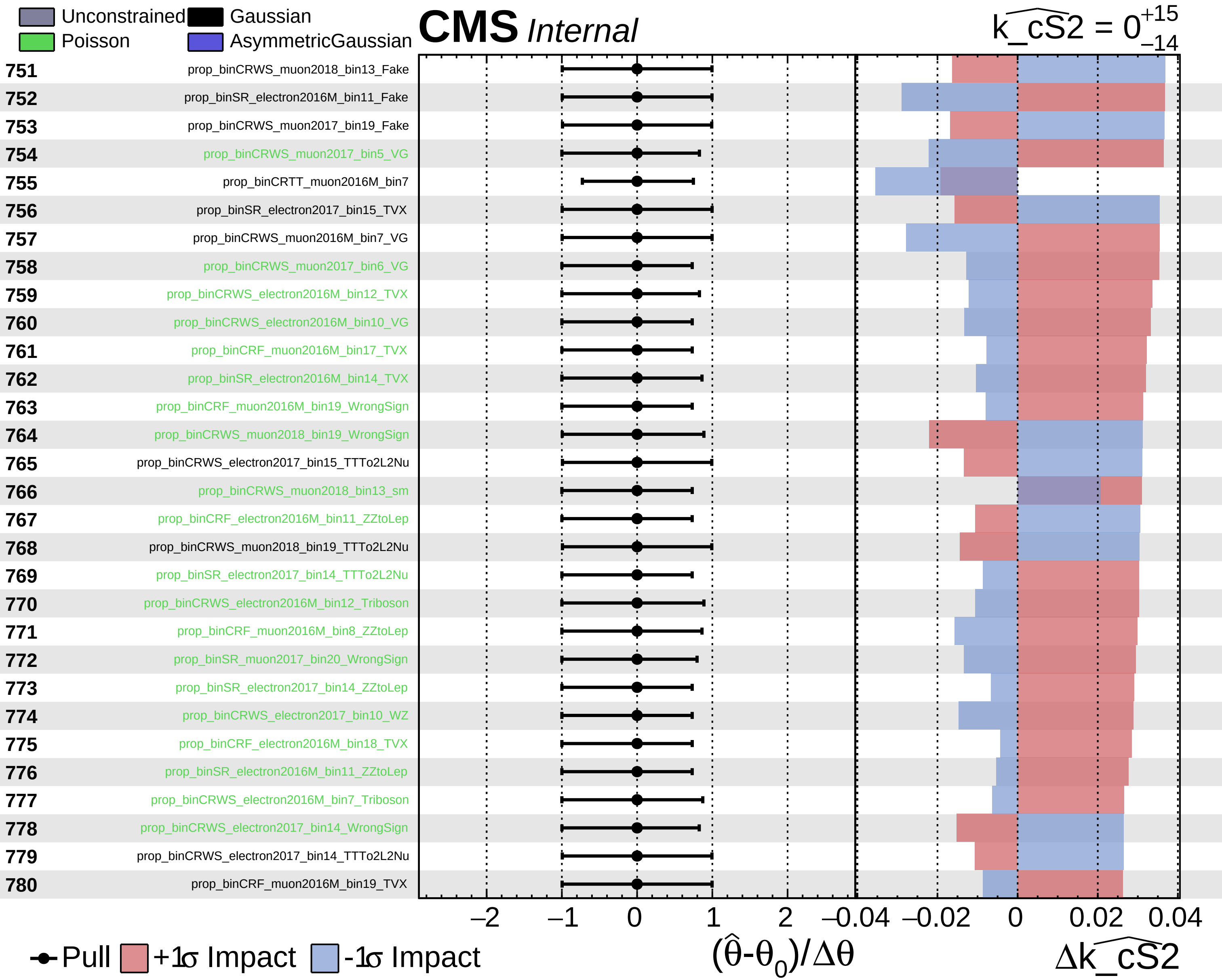
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
631 prop_binCRWS_electron2017_bin9_WZ
632 prop_binCRWS_muon2016M_bin11_WrongSign
633 prop_binCRF_electron2016M_bin20
634 prop_binCRWS_electron2018_bin13_WrongSign
635 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
636 prop_binSR_muon2016M_bin19_WrongSign
637 prop_binCRTT_muon2018_bin6
638 prop_binCRWS_electron2016M_bin16_Fake
639 prop_binCRF_electron2016M_bin12_sm
640 prop_binCRTT_electron2016M_bin9
641 prop_binSR_electron2017_bin15_TTTo2L2Nu
642 prop_binCRWS_electron2017_bin3
643 prop_binSR_electron2016M_bin14_ZZtoLep
644 prop_binCRWS_electron2018_bin5_TTTo2L2Nu
645 prop_binCRWS_muon2018_bin7_TTTo2L2Nu
646 prop_binCRWS_muon2016M_bin11_TTTo2L2Nu
647 prop_binCRWS_muon2018_bin11_WrongSign
648 prop_binCRWS_electron2018_bin13_TTTo2L2Nu
649 prop_binCRWS_muon2018_bin14_VG
650 prop_binSR_muon2018_bin11_ZZtoLep
651 prop_binCRWS_electron2016M_bin5_WZ
652 prop_binCRWS_muon2016M_bin16_Fake
653 prop_binCRWS_electron2017_bin9_TVX
654 prop_binCRWS_muon2018_bin19_Fake
655 prop_binCRWS_muon2018_bin11_WZ
656 prop_binCRWS_electron2016M_bin6_WrongSign
657 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
658 prop_binCRWS_electron2017_bin8_WrongSign
659 prop_binSR_muon2016M_bin19_TTTo2L2Nu
660 prop_binCRWS_electron2016M_bin20
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

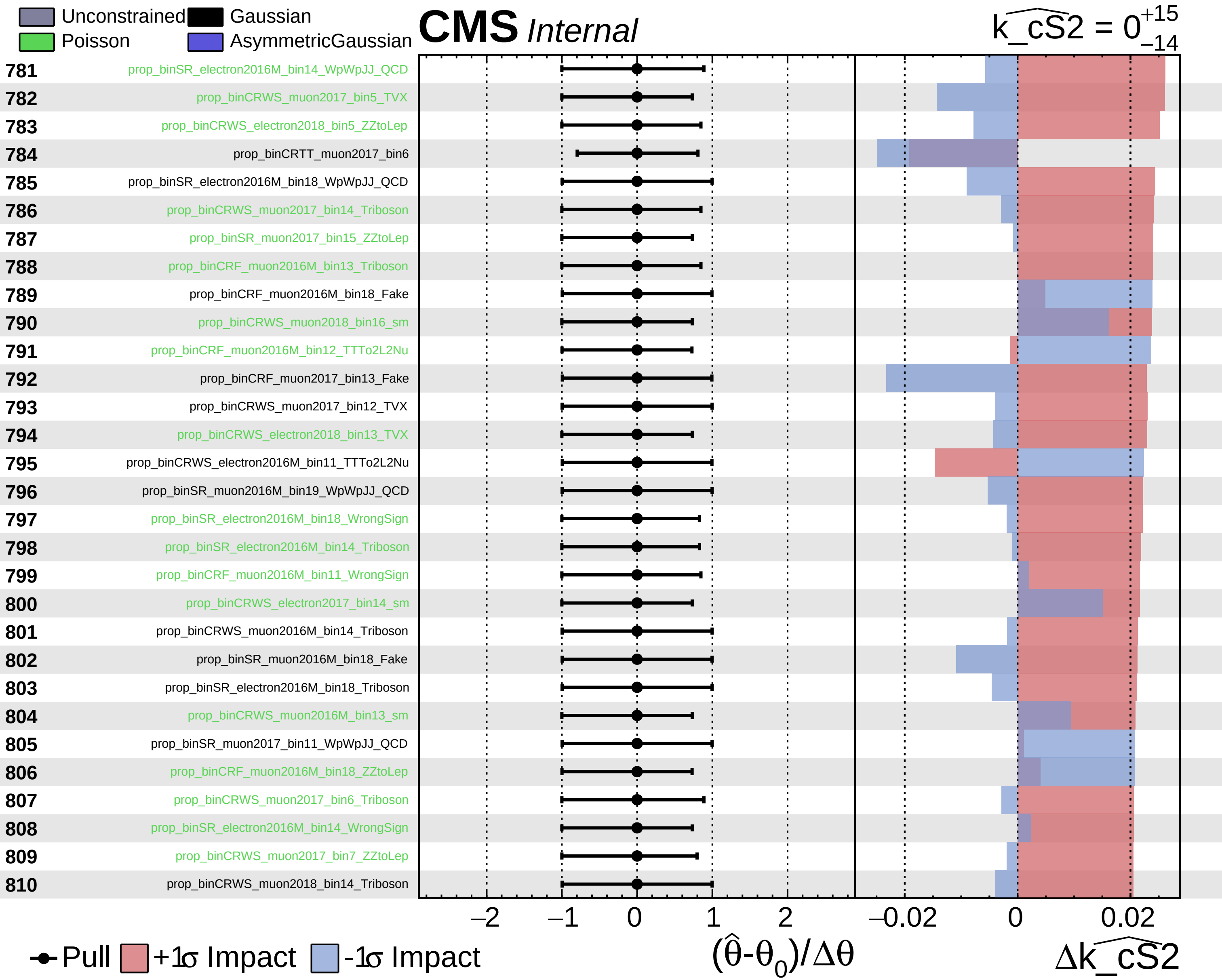


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
661 prop_binCRWS_electron2016M_bin9_WrongSign
662 prop_binCRWS_muon2018_bin16_Fake
663 prop_binCRWS_electron2017_bin8_TTTo2L2Nu
664 prop_binCRTT_electron2016M_bin10
665 prop_binCRWS_muon2018_bin11_DYJetsToLL_FxFx
666 prop_binCRWS_electron2017_bin10_WrongSign
667 prop_binCRF_electron2018_bin6_sm
668 prop_binCRF_electron2018_bin6_Fake
669 prop_binCRWS_electron2018_bin18_VG
670 prop_binCRF_electron2017_bin16_VG
671 prop_binSR_muon2017_bin20_TVX
672 prop_binCRWS_electron2016M_bin16_WrongSign
673 prop_binCRWS_electron2016M_bin10_WrongSign
674 prop_binSR_electron2018_bin15_TTTo2L2Nu
675 prop_binCRWS_muon2018_bin11_TTTo2L2Nu
676 prop_binCRTT_muon2017_bin7
677 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
678 prop_binCRF_muon2018_bin18_WrongSign
679 prop_binCRWS_electron2018_bin5_TVX
680 prop_binCRF_electron2016M_bin16_Fake
681 prop_binSR_muon2017_bin20_WpWpJJ_QCD
682 prop_binCRWS_electron2017_bin15_Fake
683 prop_binCRWS_electron2016M_bin16_TTTo2L2Nu
684 prop_binCRWS_electron2017_bin19
685 prop_binCRWS_muon2017_bin10_VG
686 prop_binCRWS_electron2017_bin10_TTTo2L2Nu
687 prop_binCRWS_muon2017_bin19_WrongSign
688 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
689 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
690 prop_binCRWS_electron2018_bin18_Fake
−2
−1
0
1
2
−0.1
−0.05
0
0.05
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

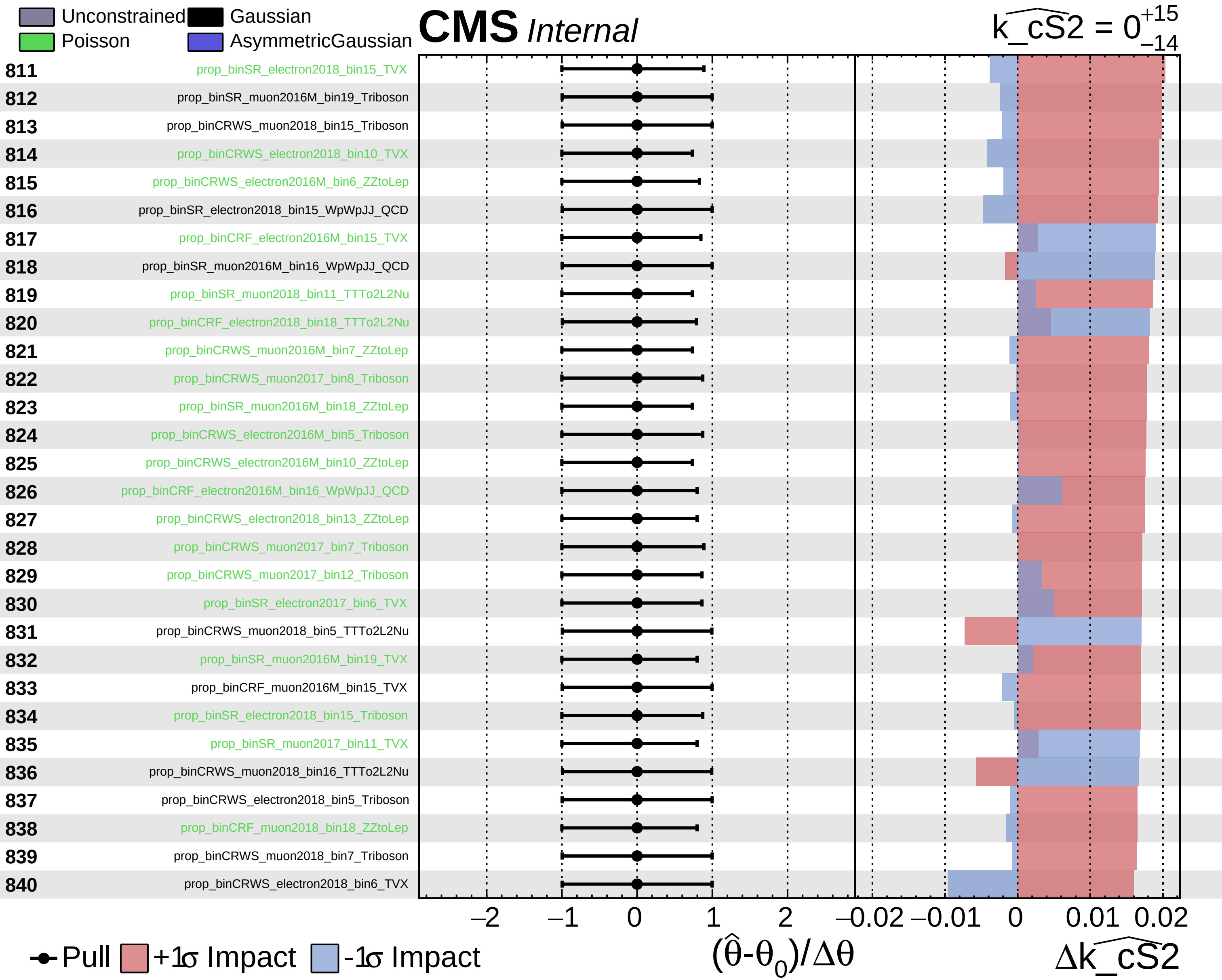

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0+15 −14
691 prop_binCRF_muon2016M_bin13_TVX
692 prop_binCRWS_muon2018_bin7_VG
693 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
694 prop_binCRWS_muon2017_bin5_WrongSign
695 prop_binCRWS_electron2017_bin13_WrongSign
696 lumi_2018
697 prop_binCRWS_electron2017_bin15_WrongSign
698 prop_binCRWS_electron2016M_bin5_VG
699 prop_binCRF_muon2018_bin18_TVX
700 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
701 prop_binCRWS_electron2017_bin8_WZ
702 prop_binCRWS_muon2017_bin7_VG
703 prop_binCRWS_electron2017_bin17
704 prop_binCRWS_electron2016M_bin6_WZ
705 prop_binCRWS_muon2016M_bin5_TTTo2L2Nu
706 prop_binCRTT_muon2018_bin8
707 prop_binCRWS_electron2016M_bin7_VG
708 prop_binCRTT_muon2016M_bin6
709 prop_binSR_muon2017_bin20_Triboson
710 prop_binCRWS_muon2017_bin16
711 prop_binCRWS_electron2017_bin20
712 prop_binCRF_muon2016M_bin17_WrongSign
713 prop_binCRWS_electron2018_bin18_WrongSign
714 prop_binCRF_electron2018_bin14_TTTo2L2Nu
715 prop_binCRF_electron2017_bin18_TVX
716 prop_binCRF_muon2016M_bin8_Triboson
717 prop_binCRWS_muon2016M_bin5_WrongSign
718 prop_binSR_muon2017_bin11_TTTo2L2Nu
719 prop_binCRWS_muon2018_bin5_WrongSign
720 prop_binCRWS_electron2018_bin10_WrongSign
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2

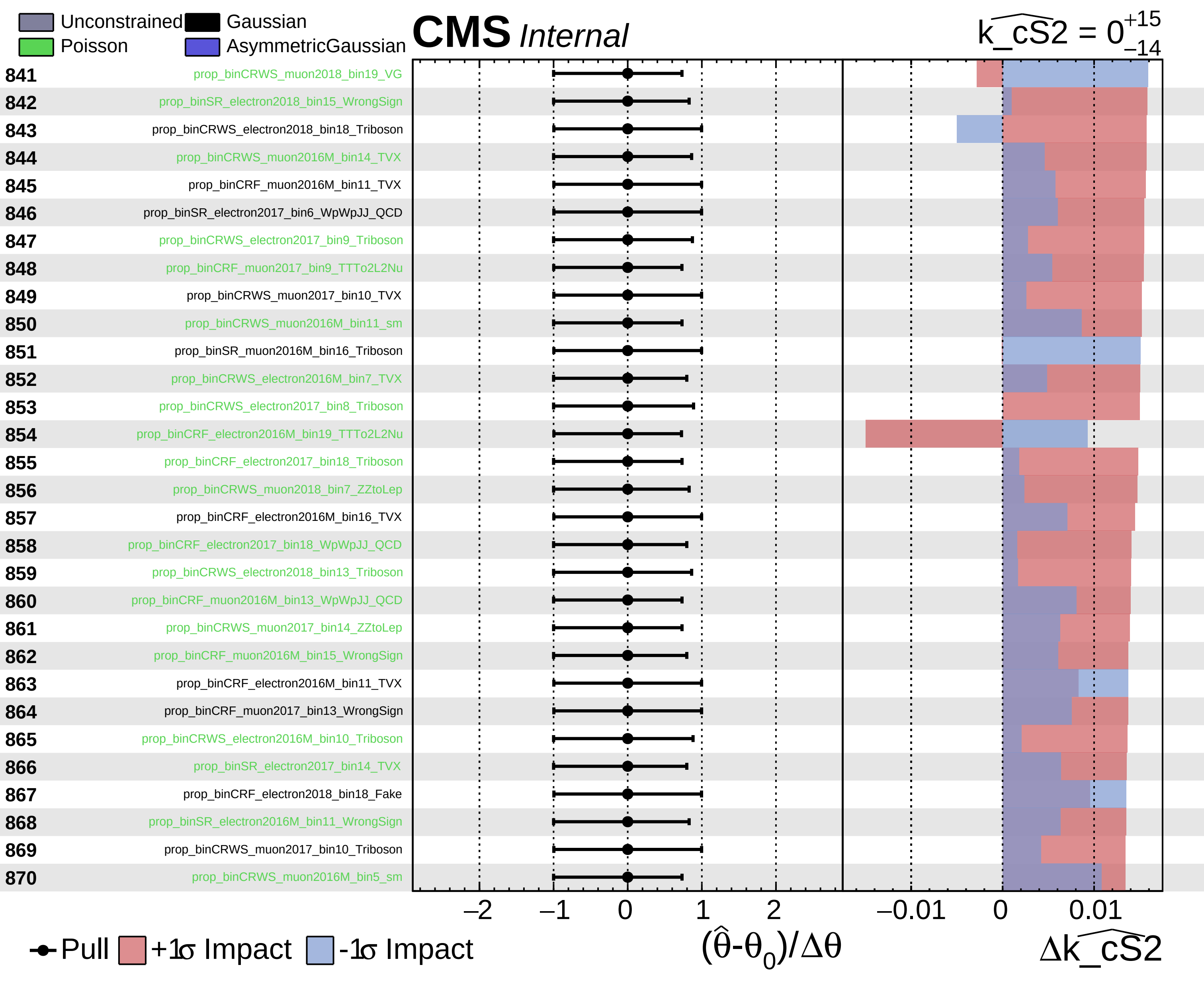

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
721 prop_binCRWS_electron2016M_bin17
722 prop_binCRWS_muon2016M_bin20
723 prop_binCRWS_muon2018_bin13_TTTo2L2Nu
724 prop_binSR_muon2016M_bin16_TTTo2L2Nu
725 prop_binCRWS_muon2017_bin7_TVX
726 prop_binCRWS_electron2018_bin10_TTTo2L2Nu
727 prop_binSR_muon2018_bin11_TVX
728 prop_binSR_muon2018_bin11_Triboson
729 prop_binSR_electron2017_bin20_WpWpJJ_QCD
730 prop_binCRWS_muon2016M_bin11_VG
731 prop_binSR_muon2018_bin11_WpWpJJ_QCD
732 prop_binSR_electron2017_bin20_TVX
733 prop_binCRWS_electron2018_bin18_TVX
734 prop_binCRF_electron2016M_bin18_WrongSign
735 prop_binCRWS_electron2016M_bin12_sm
736 prop_binCRWS_muon2017_bin8_TVX
737 prop_binCRWS_muon2018_bin16_WrongSign
738 prop_binCRWS_muon2016M_bin13_Triboson
739 prop_binSR_electron2016M_bin18_TVX
740 prop_binSR_electron2017_bin14_Fake
741 prop_binCRF_electron2017_bin18_WrongSign
742 prop_binSR_electron2016M_bin11_TTTo2L2Nu
743 prop_binSR_electron2017_bin15_WrongSign
744 prop_binCRWS_muon2018_bin13_WrongSign
745 prop_binCRWS_electron2016M_bin6_VG
746 prop_binCRF_muon2018_bin18_Triboson
747 prop_binCRWS_muon2017_bin19_TTTo2L2Nu
748 prop_binCRWS_muon2018_bin11_TVX
749 prop_binSR_electron2017_bin6_TTTo2L2Nu
750 prop_binSR_electron2017_bin15_WpWpJJ_QCD
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
751 prop_binCRWS_muon2018_bin13_Fake
752 prop_binSR_electron2016M_bin11_Fake
753 prop_binCRWS_muon2017_bin19_Fake
754 prop_binCRWS_muon2017_bin5_VG
755 prop_binCRTT_muon2016M_bin7
756 prop_binSR_electron2017_bin15_TVX
757 prop_binCRWS_muon2016M_bin7_VG
758 prop_binCRWS_muon2017_bin6_VG
759 prop_binCRWS_electron2016M_bin12_TVX
760 prop_binCRWS_electron2016M_bin10_VG
761 prop_binCRF_muon2016M_bin17_TVX
762 prop_binSR_electron2016M_bin14_TVX
763 prop_binCRF_muon2016M_bin19_WrongSign
764 prop_binCRWS_muon2018_bin19_WrongSign
765 prop_binCRWS_electron2017_bin15_TTTo2L2Nu
766 prop_binCRWS_muon2018_bin13_sm
767 prop_binCRF_electron2016M_bin11_ZZtoLep
768 prop_binCRWS_muon2018_bin19_TTTo2L2Nu
769 prop_binSR_electron2017_bin14_TTTo2L2Nu
770 prop_binCRWS_electron2016M_bin12_Triboson
771 prop_binCRF_muon2016M_bin8_ZZtoLep
772 prop_binSR_muon2017_bin20_WrongSign
773 prop_binSR_electron2017_bin14_ZZtoLep
774 prop_binCRWS_electron2017_bin10_WZ
775 prop_binCRF_electron2016M_bin18_TVX
776 prop_binSR_electron2016M_bin11_ZZtoLep
777 prop_binCRWS_electron2016M_bin7_Triboson
778 prop_binCRWS_electron2017_bin14_WrongSign
779 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
780 prop_binCRF_muon2016M_bin19_TVX
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

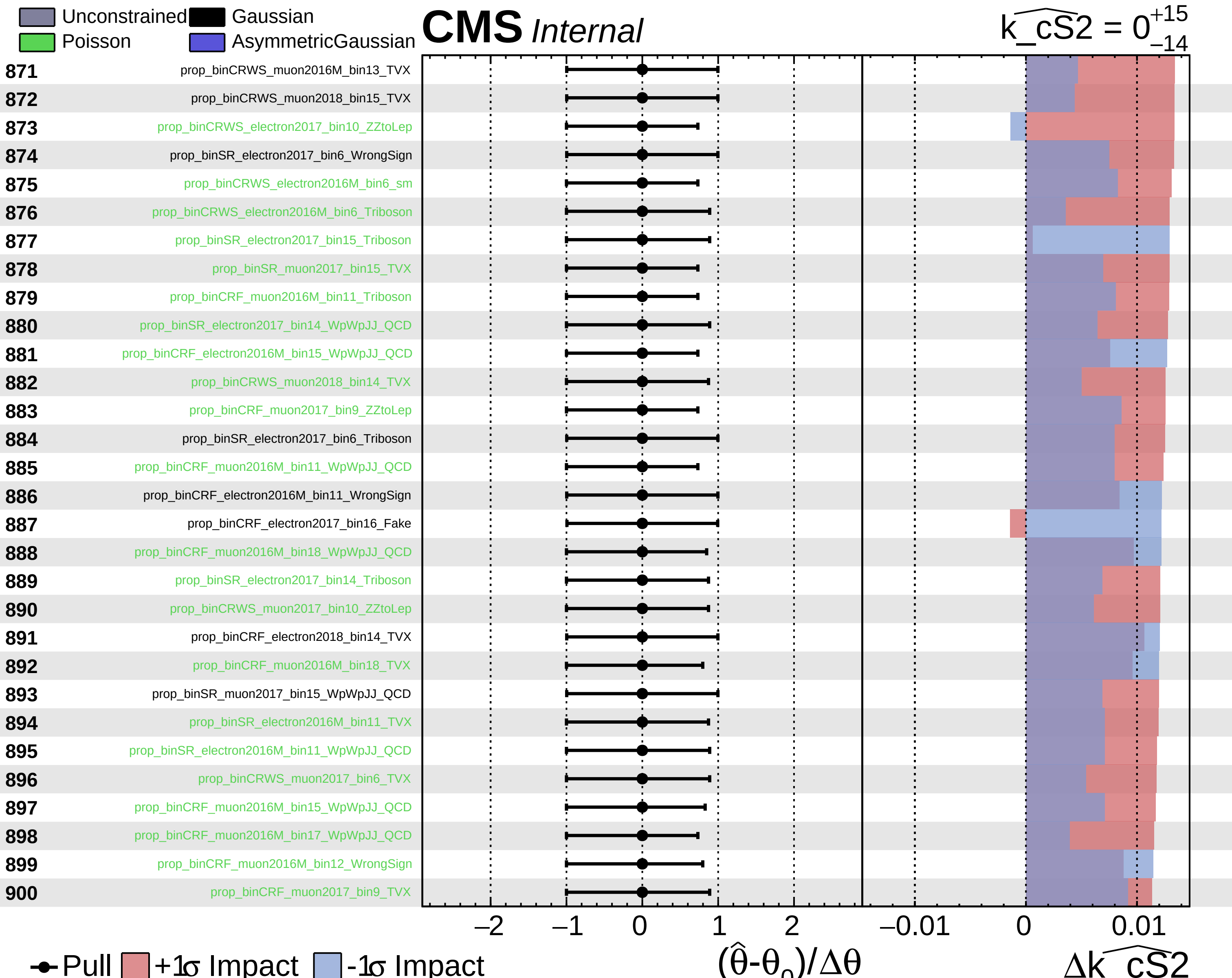


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
781 prop_binSR_electron2016M_bin14_WpWpJJ_QCD
782 prop_binCRWS_muon2017_bin5_TVX
783 prop_binCRWS_electron2018_bin5_ZZtoLep
784 prop_binCRTT_muon2017_bin6
785 prop_binSR_electron2016M_bin18_WpWpJJ_QCD
786 prop_binCRWS_muon2017_bin14_Triboson
787 prop_binSR_muon2017_bin15_ZZtoLep
788 prop_binCRF_muon2016M_bin13_Triboson
789 prop_binCRF_muon2016M_bin18_Fake
790 prop_binCRWS_muon2018_bin16_sm
791 prop_binCRF_muon2016M_bin12_TTTo2L2Nu
792 prop_binCRF_muon2017_bin13_Fake
793 prop_binCRWS_muon2017_bin12_TVX
794 prop_binCRWS_electron2018_bin13_TVX
795 prop_binCRWS_electron2016M_bin11_TTTo2L2Nu
796 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
797 prop_binSR_electron2016M_bin18_WrongSign
798 prop_binSR_electron2016M_bin14_Triboson
799 prop_binCRF_muon2016M_bin11_WrongSign
800 prop_binCRWS_electron2017_bin14_sm
801 prop_binCRWS_muon2016M_bin14_Triboson
802 prop_binSR_muon2016M_bin18_Fake
803 prop_binSR_electron2016M_bin18_Triboson
804 prop_binCRWS_muon2016M_bin13_sm
805 prop_binSR_muon2017_bin11_WpWpJJ_QCD
806 prop_binCRF_muon2016M_bin18_ZZtoLep
807 prop_binCRWS_muon2017_bin6_Triboson
808 prop_binSR_electron2016M_bin14_WrongSign
809 prop_binCRWS_muon2017_bin7_ZZtoLep
810 prop_binCRWS_muon2018_bin14_Triboson
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

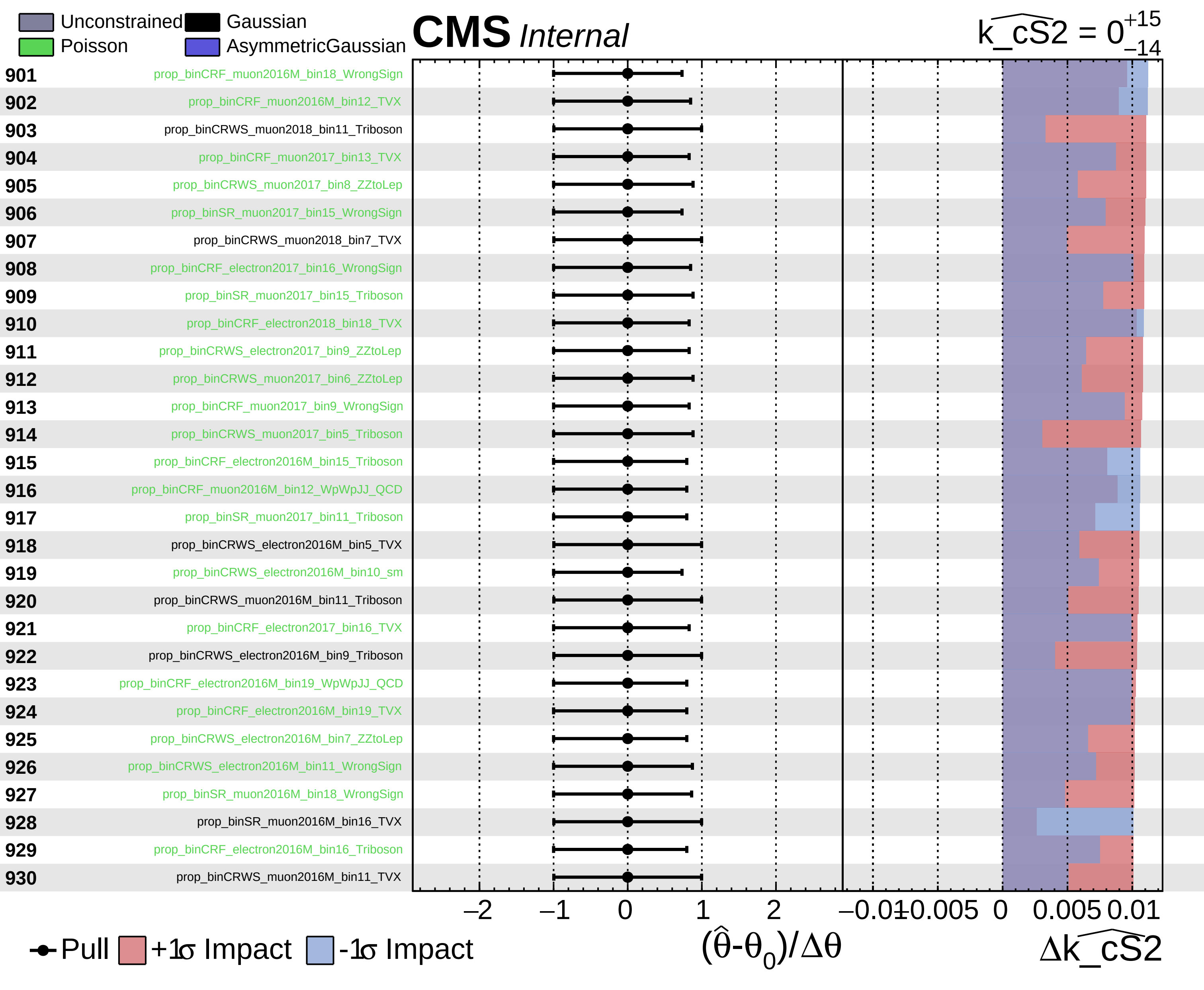

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
811 prop_binSR_electron2018_bin15_TVX
812 prop_binSR_muon2016M_bin19_Triboson
813 prop_binCRWS_muon2018_bin15_Triboson
814 prop_binCRWS_electron2018_bin10_TVX
815 prop_binCRWS_electron2016M_bin6_ZZtoLep
816 prop_binSR_electron2018_bin15_WpWpJJ_QCD
817 prop_binCRF_electron2016M_bin15_TVX
818 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
819 prop_binSR_muon2018_bin11_TTTo2L2Nu
820 prop_binCRF_electron2018_bin18_TTTo2L2Nu
821 prop_binCRWS_muon2016M_bin7_ZZtoLep
822 prop_binCRWS_muon2017_bin8_Triboson
823 prop_binSR_muon2016M_bin18_ZZtoLep
824 prop_binCRWS_electron2016M_bin5_Triboson
825 prop_binCRWS_electron2016M_bin10_ZZtoLep
826 prop_binCRF_electron2016M_bin16_WpWpJJ_QCD
827 prop_binCRWS_electron2018_bin13_ZZtoLep
828 prop_binCRWS_muon2017_bin7_Triboson
829 prop_binCRWS_muon2017_bin12_Triboson
830 prop_binSR_electron2017_bin6_TVX
831 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
832 prop_binSR_muon2016M_bin19_TVX
833 prop_binCRF_muon2016M_bin15_TVX
834 prop_binSR_electron2018_bin15_Triboson
835 prop_binSR_muon2017_bin11_TVX
836 prop_binCRWS_muon2018_bin16_TTTo2L2Nu
837 prop_binCRWS_electron2018_bin5_Triboson
838 prop_binCRF_muon2018_bin18_ZZtoLep
839 prop_binCRWS_muon2018_bin7_Triboson
840 prop_binCRWS_electron2018_bin6_TVX
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



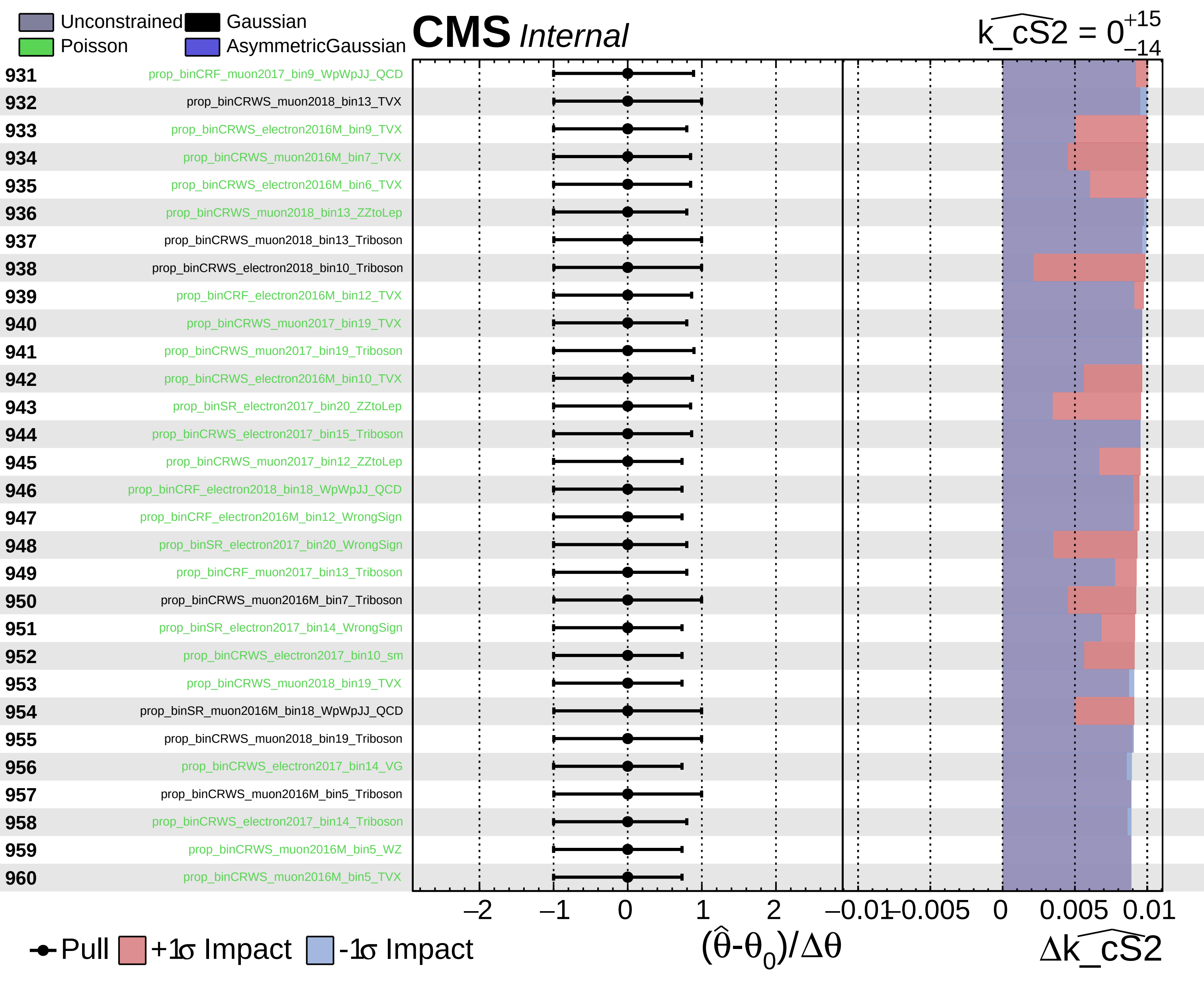
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
841 prop_binCRWS_muon2018_bin19_VG
842 prop_binSR_electron2018_bin15_WrongSign
843 prop_binCRWS_electron2018_bin18_Triboson
844 prop_binCRWS_muon2016M_bin14_TVX
845 prop_binCRF_muon2016M_bin11_TVX
846 prop_binSR_electron2017_bin6_WpWpJJ_QCD
847 prop_binCRWS_electron2017_bin9_Triboson
848 prop_binCRF_muon2017_bin9_TTTo2L2Nu
849 prop_binCRWS_muon2017_bin10_TVX
850 prop_binCRWS_muon2016M_bin11_sm
851 prop_binSR_muon2016M_bin16_Triboson
852 prop_binCRWS_electron2016M_bin7_TVX
853 prop_binCRWS_electron2017_bin8_Triboson
854 prop_binCRF_electron2016M_bin19_TTTo2L2Nu
855 prop_binCRF_electron2017_bin18_Triboson
856 prop_binCRWS_muon2018_bin7_ZZtoLep
857 prop_binCRF_electron2016M_bin16_TVX
858 prop_binCRF_electron2017_bin18_WpWpJJ_QCD
859 prop_binCRWS_electron2018_bin13_Triboson
860 prop_binCRF_muon2016M_bin13_WpWpJJ_QCD
861 prop_binCRWS_muon2017_bin14_ZZtoLep
862 prop_binCRF_muon2016M_bin15_WrongSign
863 prop_binCRF_electron2016M_bin11_TVX
864 prop_binCRF_muon2017_bin13_WrongSign
865 prop_binCRWS_electron2016M_bin10_Triboson
866 prop_binSR_electron2017_bin14_TVX
867 prop_binCRF_electron2018_bin18_Fake
868 prop_binSR_electron2016M_bin11_WrongSign
869 prop_binCRWS_muon2017_bin10_Triboson
870 prop_binCRWS_muon2016M_bin5_sm
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

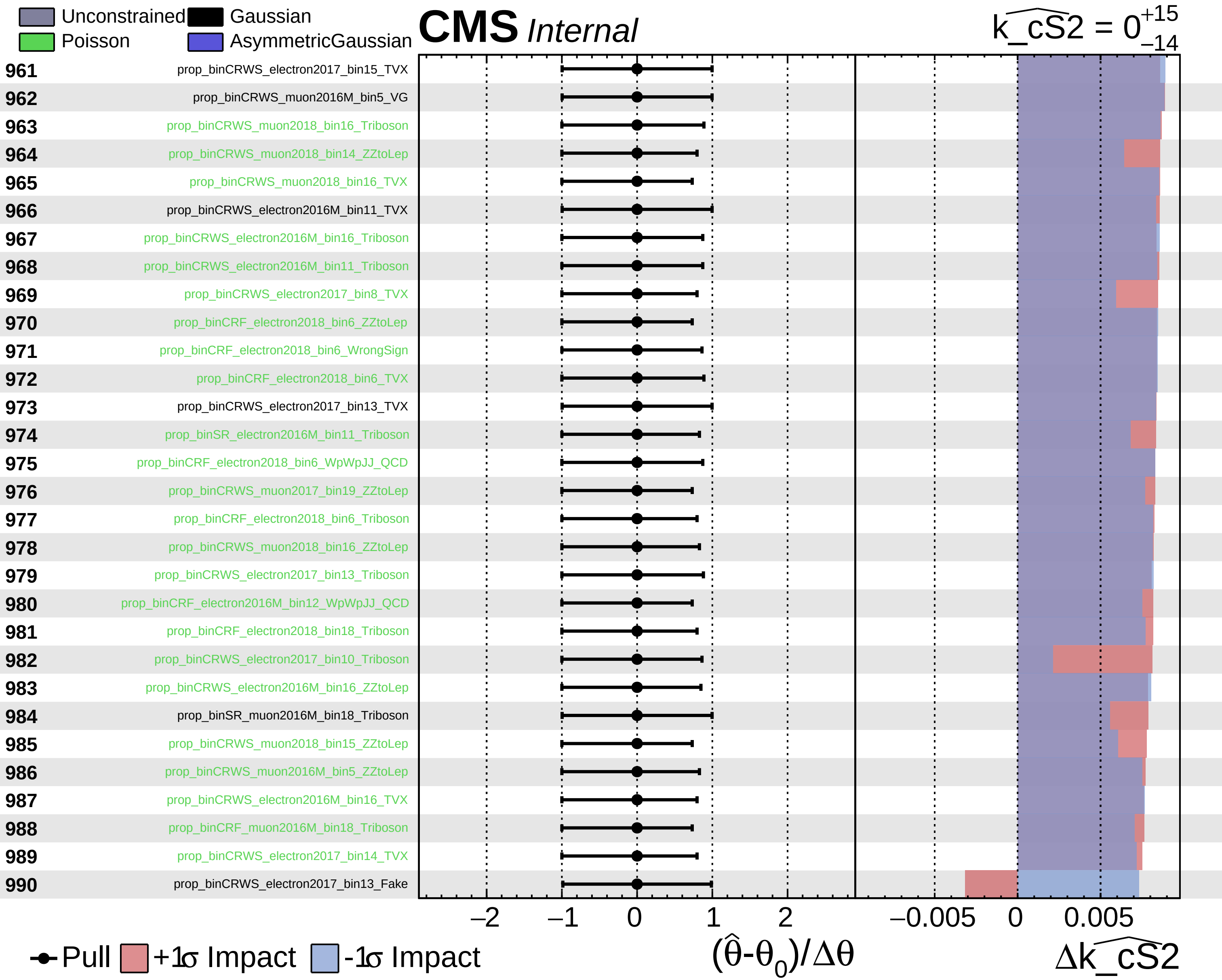


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0+15−14
871 prop_binCRWS_muon2016M_bin13_TVX
872 prop_binCRWS_muon2018_bin15_TVX
873 prop_binCRWS_electron2017_bin10_ZZtoLep
874 prop_binSR_electron2017_bin6_WrongSign
875 prop_binCRWS_electron2016M_bin6_sm
876 prop_binCRWS_electron2016M_bin6_Triboson
877 prop_binSR_electron2017_bin15_Triboson
878 prop_binSR_muon2017_bin15_TVX
879 prop_binCRF_muon2016M_bin11_Triboson
880 prop_binSR_electron2017_bin14_WpWpJJ_QCD
881 prop_binCRF_electron2016M_bin15_WpWpJJ_QCD
882 prop_binCRWS_muon2018_bin14_TVX
883 prop_binCRF_muon2017_bin9_ZZtoLep
884 prop_binSR_electron2017_bin6_Triboson
885 prop_binCRF_muon2016M_bin11_WpWpJJ_QCD
886 prop_binCRF_electron2016M_bin11_WrongSign
887 prop_binCRF_electron2017_bin16_Fake
888 prop_binCRF_muon2016M_bin18_WpWpJJ_QCD
889 prop_binSR_electron2017_bin14_Triboson
890 prop_binCRWS_muon2017_bin10_ZZtoLep
891 prop_binCRF_electron2018_bin14_TVX
892 prop_binCRF_muon2016M_bin18_TVX
893 prop_binSR_muon2017_bin15_WpWpJJ_QCD
894 prop_binSR_electron2016M_bin11_TVX
895 prop_binSR_electron2016M_bin11_WpWpJJ_QCD
896 prop_binCRWS_muon2017_bin6_TVX
897 prop_binCRF_muon2016M_bin15_WpWpJJ_QCD
898 prop_binCRF_muon2016M_bin17_WpWpJJ_QCD
899 prop_binCRF_muon2016M_bin12_WrongSign
900 prop_binCRF_muon2017_bin9_TVX
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2

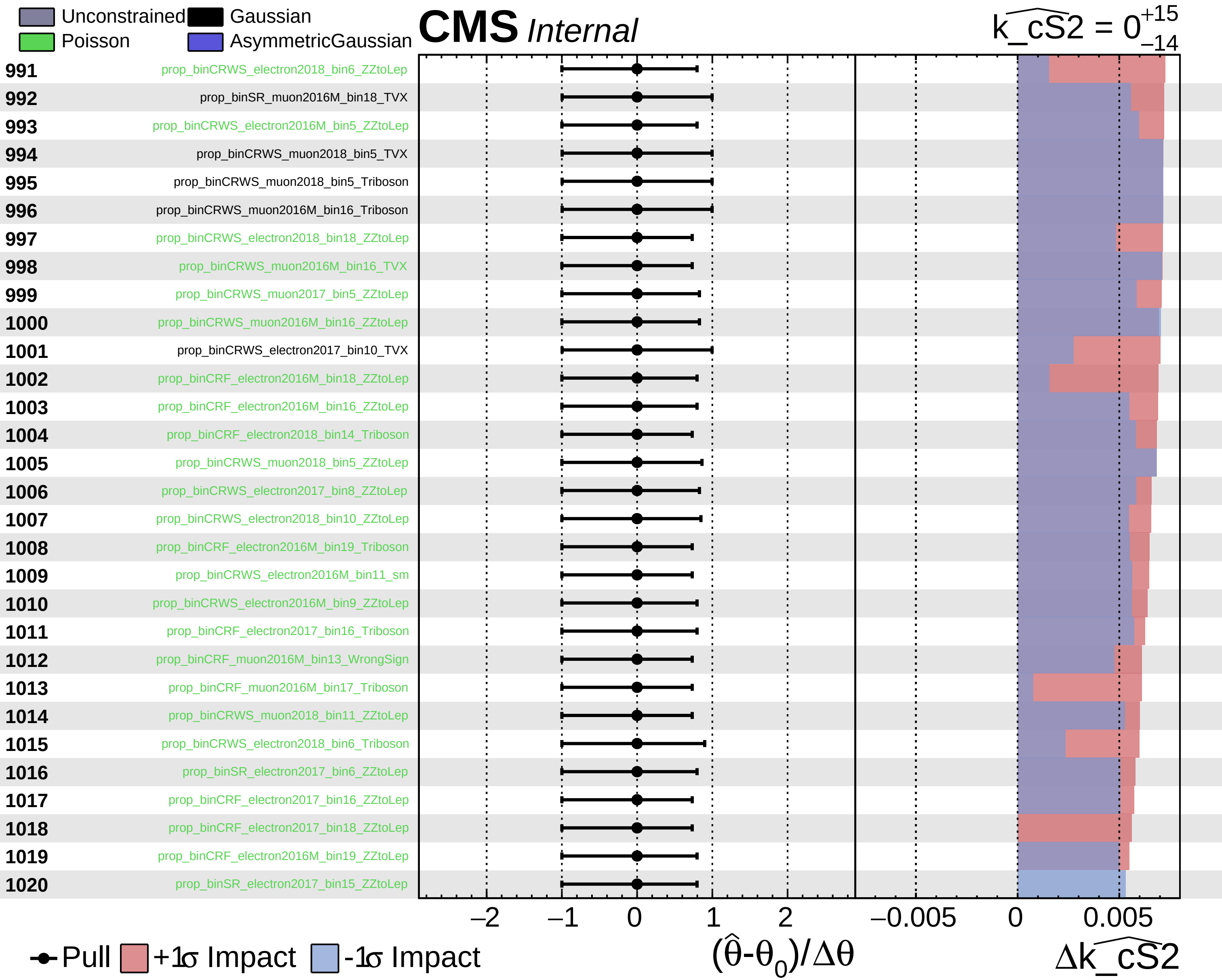


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
901 prop_binCRF_muon2016M_bin18_WrongSign
902 prop_binCRF_muon2016M_bin12_TVX
903 prop_binCRWS_muon2018_bin11_Triboson
904 prop_binCRF_muon2017_bin13_TVX
905 prop_binCRWS_muon2017_bin8_ZZtoLep
906 prop_binSR_muon2017_bin15_WrongSign
907 prop_binCRWS_muon2018_bin7_TVX
908 prop_binCRF_electron2017_bin16_WrongSign
909 prop_binSR_muon2017_bin15_Triboson
910 prop_binCRF_electron2018_bin18_TVX
911 prop_binCRWS_electron2017_bin9_ZZtoLep
912 prop_binCRWS_muon2017_bin6_ZZtoLep
913 prop_binCRF_muon2017_bin9_WrongSign
914 prop_binCRWS_muon2017_bin5_Triboson
915 prop_binCRF_electron2016M_bin15_Triboson
916 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
917 prop_binSR_muon2017_bin11_Triboson
918 prop_binCRWS_electron2016M_bin5_TVX
919 prop_binCRWS_electron2016M_bin10_sm
920 prop_binCRWS_muon2016M_bin11_Triboson
921 prop_binCRF_electron2017_bin16_TVX
922 prop_binCRWS_electron2016M_bin9_Triboson
923 prop_binCRF_electron2016M_bin19_WpWpJJ_QCD
924 prop_binCRF_electron2016M_bin19_TVX
925 prop_binCRWS_electron2016M_bin7_ZZtoLep
926 prop_binCRWS_electron2016M_bin11_WrongSign
927 prop_binSR_muon2016M_bin18_WrongSign
928 prop_binSR_muon2016M_bin16_TVX
929 prop_binCRF_electron2016M_bin16_Triboson
930 prop_binCRWS_muon2016M_bin11_TVX
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cS2

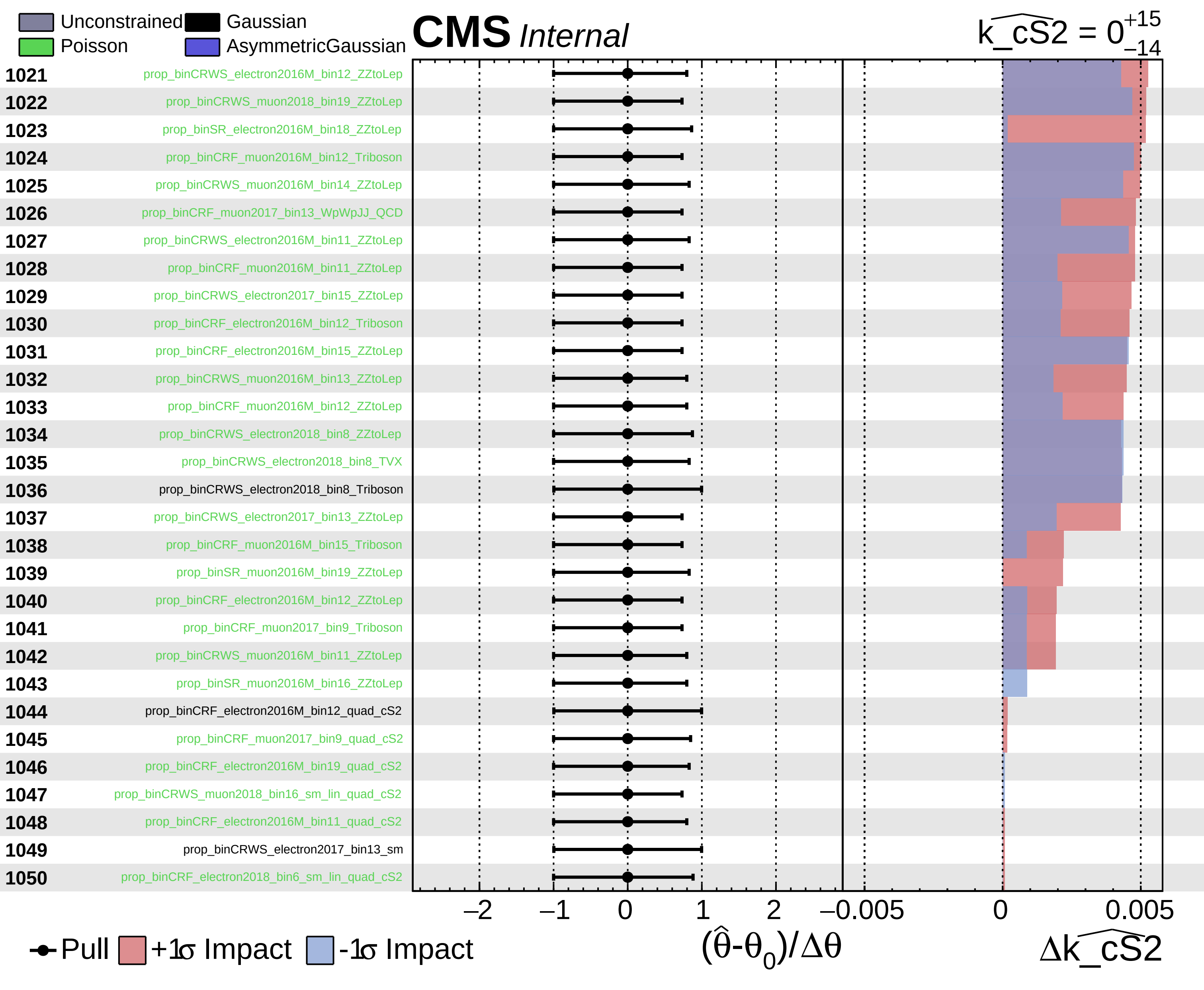

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
931 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
932 prop_binCRWS_muon2018_bin13_TVX
933 prop_binCRWS_electron2016M_bin9_TVX
934 prop_binCRWS_muon2016M_bin7_TVX
935 prop_binCRWS_electron2016M_bin6_TVX
936 prop_binCRWS_muon2018_bin13_ZZtoLep
937 prop_binCRWS_muon2018_bin13_Triboson
938 prop_binCRWS_electron2018_bin10_Triboson
939 prop_binCRF_electron2016M_bin12_TVX
940 prop_binCRWS_muon2017_bin19_TVX
941 prop_binCRWS_muon2017_bin19_Triboson
942 prop_binCRWS_electron2016M_bin10_TVX
943 prop_binSR_electron2017_bin20_ZZtoLep
944 prop_binCRWS_electron2017_bin15_Triboson
945 prop_binCRWS_muon2017_bin12_ZZtoLep
946 prop_binCRF_electron2018_bin18_WpWpJJ_QCD
947 prop_binCRF_electron2016M_bin12_WrongSign
948 prop_binSR_electron2017_bin20_WrongSign
949 prop_binCRF_muon2017_bin13_Triboson
950 prop_binCRWS_muon2016M_bin7_Triboson
951 prop_binSR_electron2017_bin14_WrongSign
952 prop_binCRWS_electron2017_bin10_sm
953 prop_binCRWS_muon2018_bin19_TVX
954 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
955 prop_binCRWS_muon2018_bin19_Triboson
956 prop_binCRWS_electron2017_bin14_VG
957 prop_binCRWS_muon2016M_bin5_Triboson
958 prop_binCRWS_electron2017_bin14_Triboson
959 prop_binCRWS_muon2016M_bin5_WZ
960 prop_binCRWS_muon2016M_bin5_TVX
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



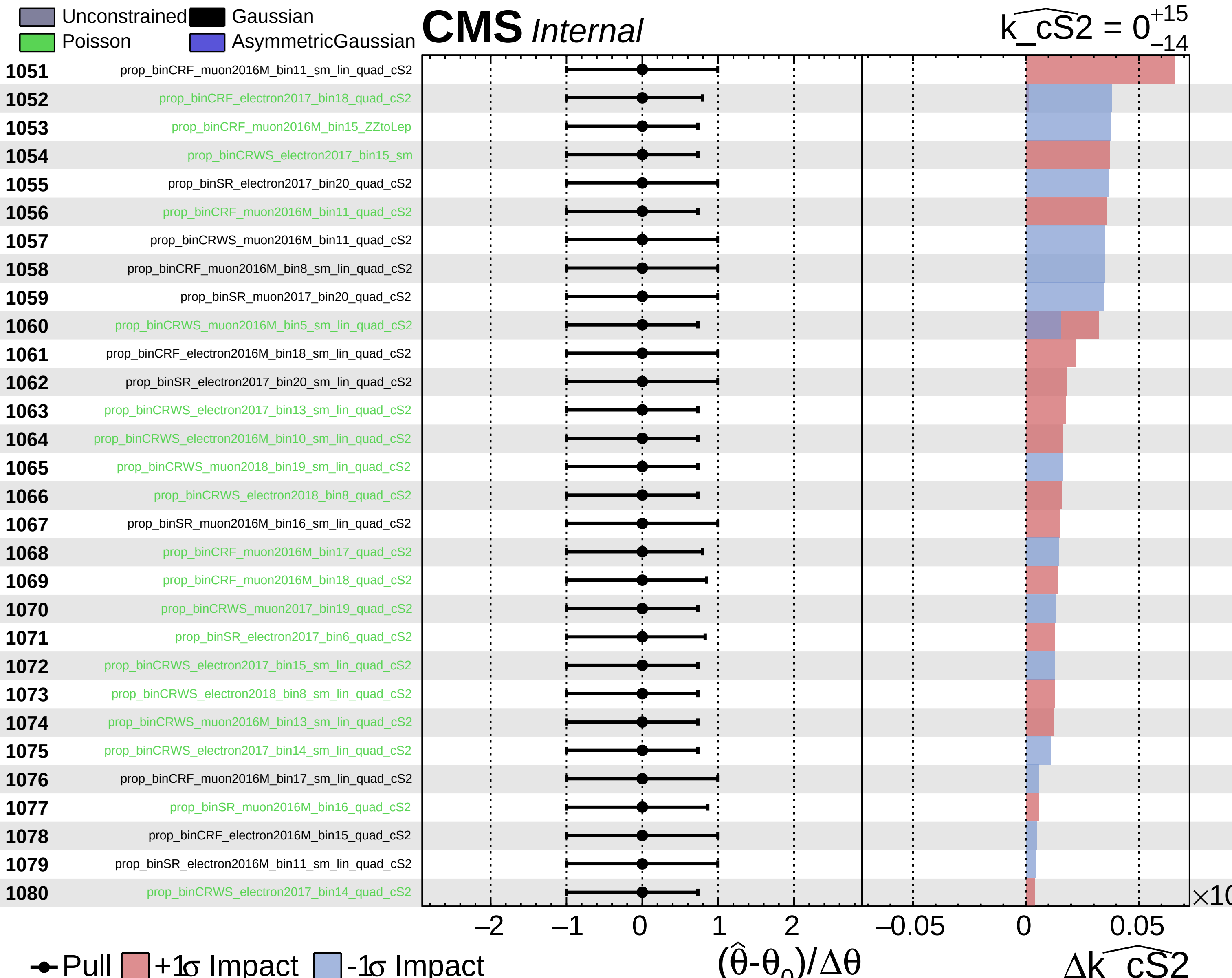
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
961 prop_binCRWS_electron2017_bin15_TVX
962 prop_binCRWS_muon2016M_bin5_VG
963 prop_binCRWS_muon2018_bin16_Triboson
964 prop_binCRWS_muon2018_bin14_ZZtoLep
965 prop_binCRWS_muon2018_bin16_TVX
966 prop_binCRWS_electron2016M_bin11_TVX
967 prop_binCRWS_electron2016M_bin16_Triboson
968 prop_binCRWS_electron2016M_bin11_Triboson
969 prop_binCRWS_electron2017_bin8_TVX
970 prop_binCRF_electron2018_bin6_ZZtoLep
971 prop_binCRF_electron2018_bin6_WrongSign
972 prop_binCRF_electron2018_bin6_TVX
973 prop_binCRWS_electron2017_bin13_TVX
974 prop_binSR_electron2016M_bin11_Triboson
975 prop_binCRF_electron2018_bin6_WpWpJJ_QCD
976 prop_binCRWS_muon2017_bin19_ZZtoLep
977 prop_binCRF_electron2018_bin6_Triboson
978 prop_binCRWS_muon2018_bin16_ZZtoLep
979 prop_binCRWS_electron2017_bin13_Triboson
980 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
981 prop_binCRF_electron2018_bin18_Triboson
982 prop_binCRWS_electron2017_bin10_Triboson
983 prop_binCRWS_electron2016M_bin16_ZZtoLep
984 prop_binSR_muon2016M_bin18_Triboson
985 prop_binCRWS_muon2018_bin15_ZZtoLep
986 prop_binCRWS_muon2016M_bin5_ZZtoLep
987 prop_binCRWS_electron2016M_bin16_TVX
988 prop_binCRF_muon2016M_bin18_Triboson
989 prop_binCRWS_electron2017_bin14_TVX
990 prop_binCRWS_electron2017_bin13_Fake
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



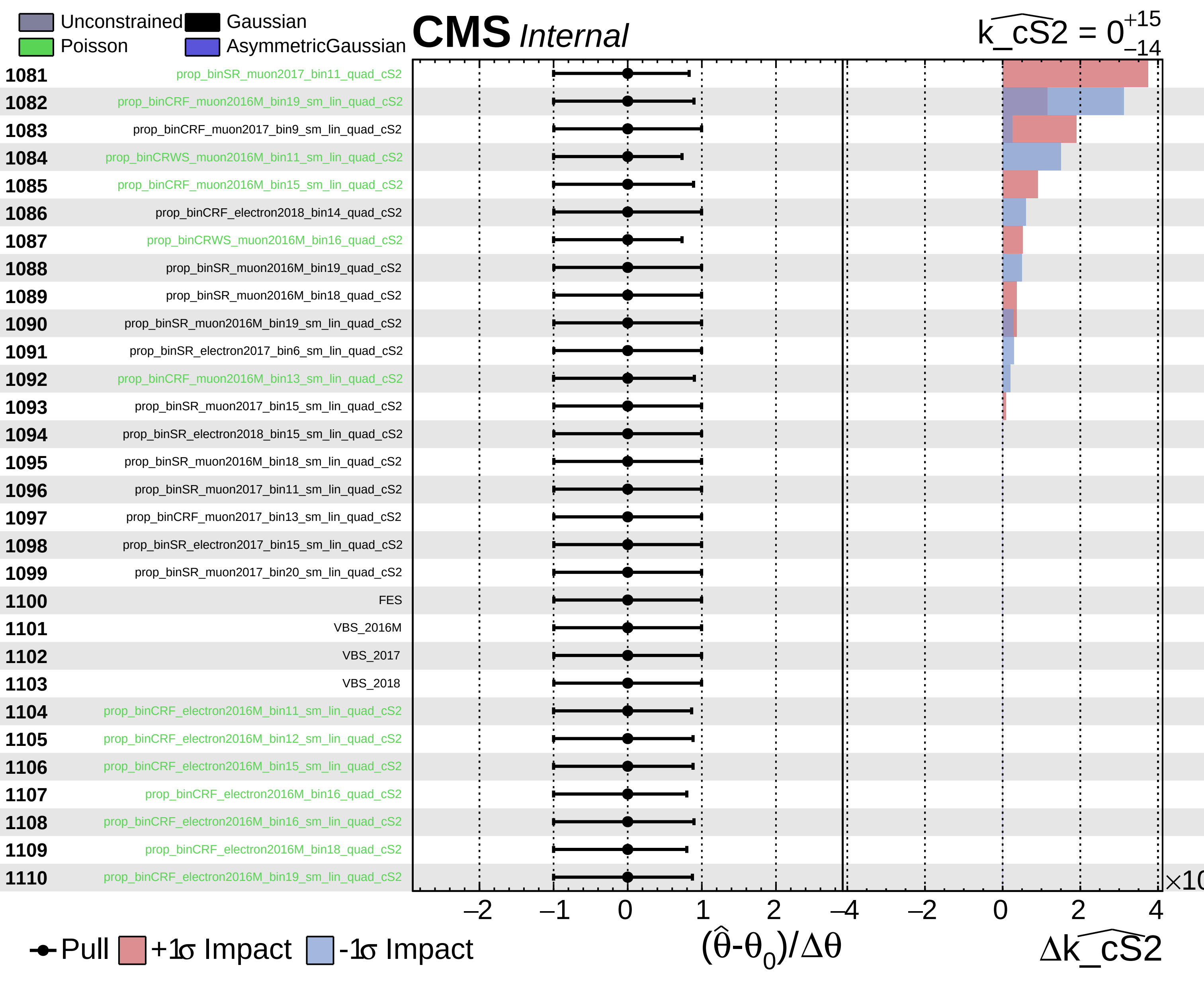
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
991 prop_binCRWS_electron2018_bin6_ZZtoLep
992 prop_binSR_muon2016M_bin18_TVX
993 prop_binCRWS_electron2016M_bin5_ZZtoLep
994 prop_binCRWS_muon2018_bin5_TVX
995 prop_binCRWS_muon2018_bin5_Triboson
996 prop_binCRWS_muon2016M_bin16_Triboson
997 prop_binCRWS_electron2018_bin18_ZZtoLep
998 prop_binCRWS_muon2016M_bin16_TVX
999 prop_binCRWS_muon2017_bin5_ZZtoLep
1000 prop_binCRWS_muon2016M_bin16_ZZtoLep
1001 prop_binCRWS_electron2017_bin10_TVX
1002 prop_binCRF_electron2016M_bin18_ZZtoLep
1003 prop_binCRF_electron2016M_bin16_ZZtoLep
1004 prop_binCRF_electron2018_bin14_Triboson
1005 prop_binCRWS_muon2018_bin5_ZZtoLep
1006 prop_binCRWS_electron2017_bin8_ZZtoLep
1007 prop_binCRWS_electron2018_bin10_ZZtoLep
1008 prop_binCRF_electron2016M_bin19_Triboson
1009 prop_binCRWS_electron2016M_bin11_sm
1010 prop_binCRWS_electron2016M_bin9_ZZtoLep
1011 prop_binCRF_electron2017_bin16_Triboson
1012 prop_binCRF_muon2016M_bin13_WrongSign
1013 prop_binCRF_muon2016M_bin17_Triboson
1014 prop_binCRWS_muon2018_bin11_ZZtoLep
1015 prop_binCRWS_electron2018_bin6_Triboson
1016 prop_binSR_electron2017_bin6_ZZtoLep
1017 prop_binCRF_electron2017_bin16_ZZtoLep
1018 prop_binCRF_electron2017_bin18_ZZtoLep
1019 prop_binCRF_electron2016M_bin19_ZZtoLep
1020 prop_binSR_electron2017_bin15_ZZtoLep
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2



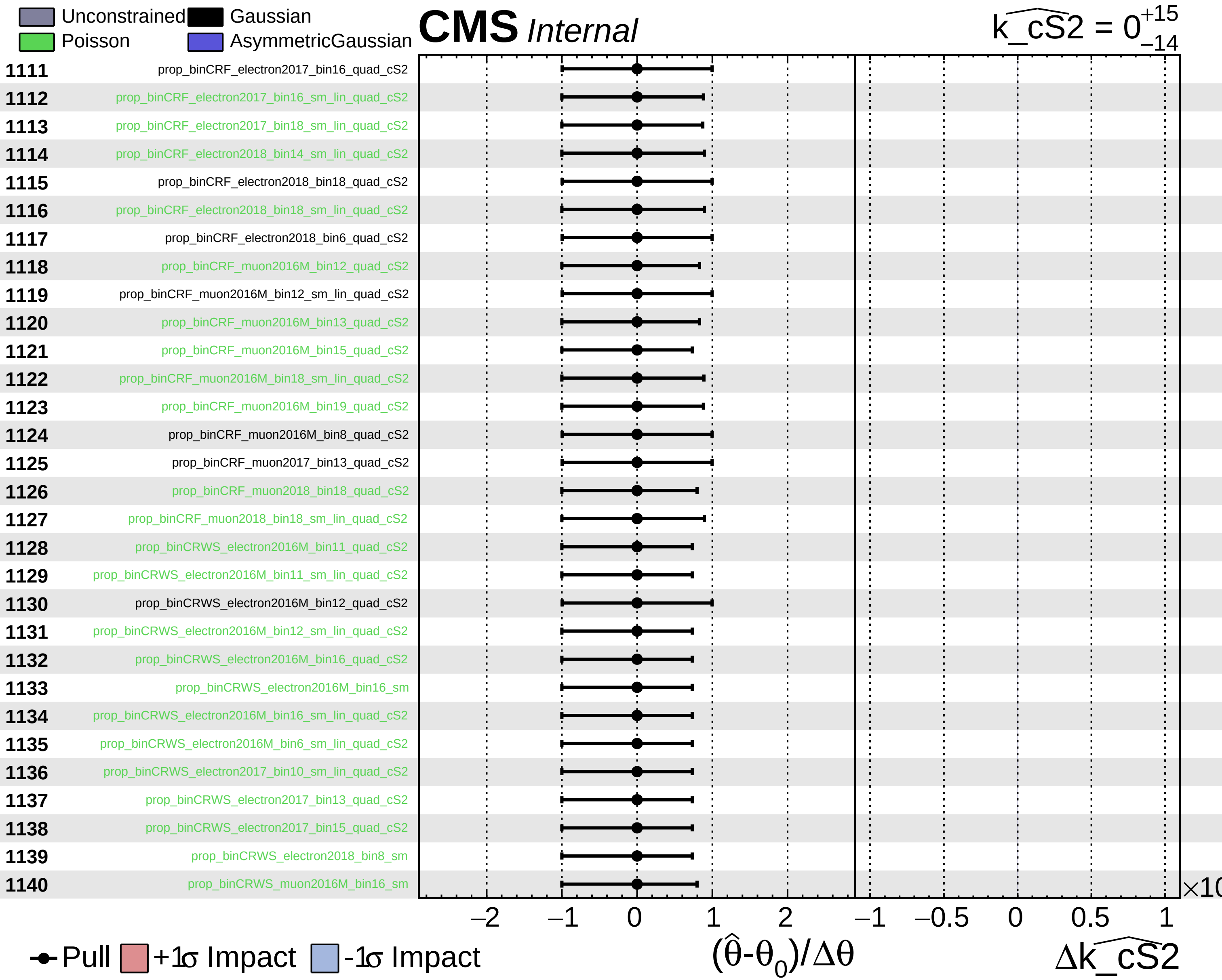
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
1021 prop_binCRWS_electron2016M_bin12_ZZtoLep
1022 prop_binCRWS_muon2018_bin19_ZZtoLep
1023 prop_binSR_electron2016M_bin18_ZZtoLep
1024 prop_binCRF_muon2016M_bin12_Triboson
1025 prop_binCRWS_muon2016M_bin14_ZZtoLep
1026 prop_binCRF_muon2017_bin13_WpWpJJ_QCD
1027 prop_binCRWS_electron2016M_bin11_ZZtoLep
1028 prop_binCRF_muon2016M_bin11_ZZtoLep
1029 prop_binCRWS_electron2017_bin15_ZZtoLep
1030 prop_binCRF_electron2016M_bin12_Triboson
1031 prop_binCRF_electron2016M_bin15_ZZtoLep
1032 prop_binCRWS_muon2016M_bin13_ZZtoLep
1033 prop_binCRF_muon2016M_bin12_ZZtoLep
1034 prop_binCRWS_electron2018_bin8_ZZtoLep
1035 prop_binCRWS_electron2018_bin8_TVX
1036 prop_binCRWS_electron2018_bin8_Triboson
1037 prop_binCRWS_electron2017_bin13_ZZtoLep
1038 prop_binCRF_muon2016M_bin15_Triboson
1039 prop_binSR_muon2016M_bin19_ZZtoLep
1040 prop_binCRF_electron2016M_bin12_ZZtoLep
1041 prop_binCRF_muon2017_bin9_Triboson
1042 prop_binCRWS_muon2016M_bin11_ZZtoLep
1043 prop_binSR_muon2016M_bin16_ZZtoLep
1044 prop_binCRF_electron2016M_bin12_quad_cS2
1045 prop_binCRF_muon2017_bin9_quad_cS2
1046 prop_binCRF_electron2016M_bin19_quad_cS2
1047 prop_binCRWS_muon2018_bin16_sm_lin_quad_cS2
1048 prop_binCRF_electron2016M_bin11_quad_cS2
1049 prop_binCRWS_electron2017_bin13_sm
1050 prop_binCRF_electron2018_bin6_sm_lin_quad_cS2
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2



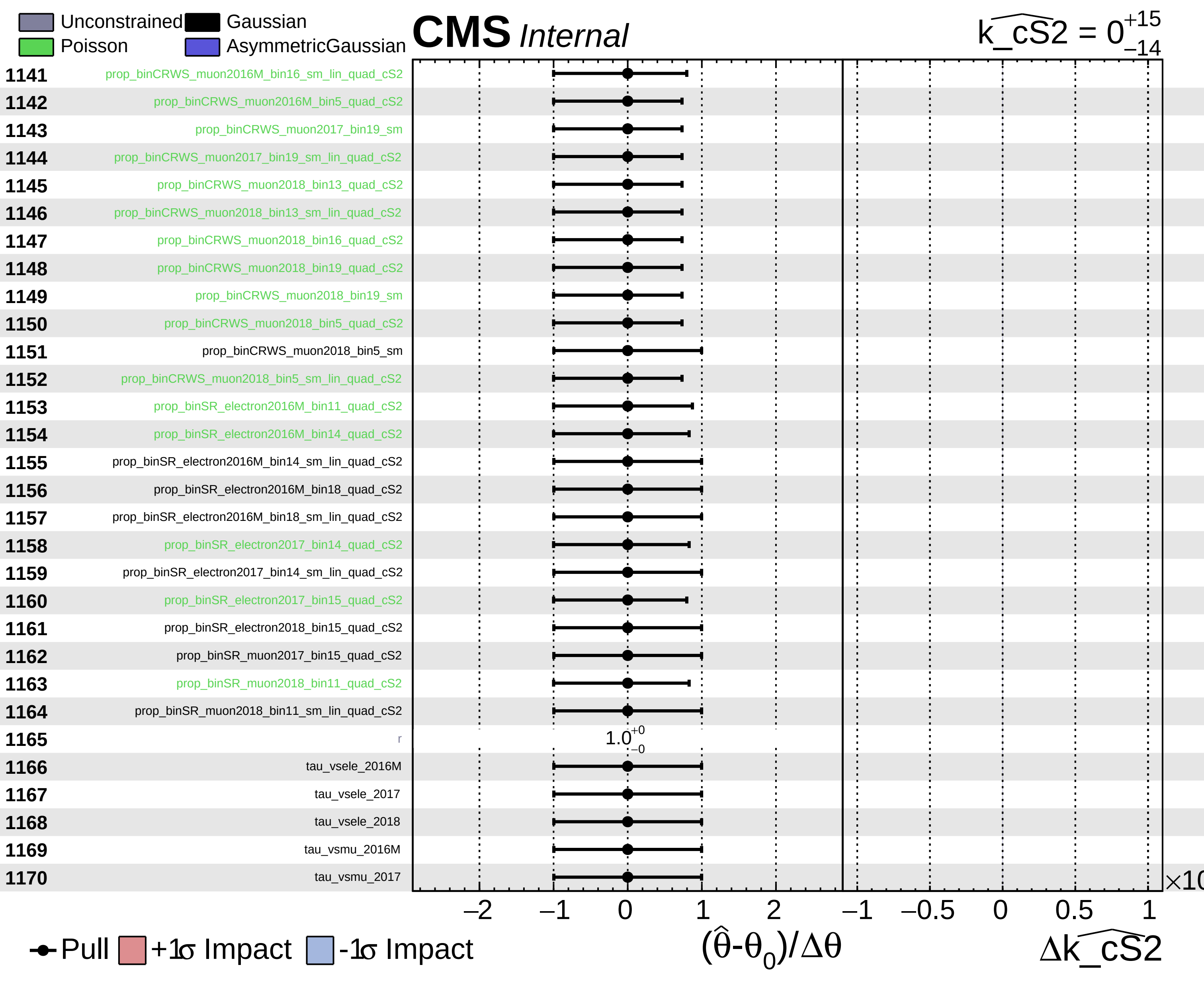
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0+15 −14
1051 prop_binCRF_muon2016M_bin11_sm_lin_quad_cS2
1052 prop_binCRF_electron2017_bin18_quad_cS2
1053 prop_binCRF_muon2016M_bin15_ZZtoLep
1054 prop_binCRWS_electron2017_bin15_sm
1055 prop_binSR_electron2017_bin20_quad_cS2
1056 prop_binCRF_muon2016M_bin11_quad_cS2
1057 prop_binCRWS_muon2016M_bin11_quad_cS2
1058 prop_binCRF_muon2016M_bin8_sm_lin_quad_cS2
1059 prop_binSR_muon2017_bin20_quad_cS2
1060 prop_binCRWS_muon2016M_bin5_sm_lin_quad_cS2
1061 prop_binCRF_electron2016M_bin18_sm_lin_quad_cS2
1062 prop_binSR_electron2017_bin20_sm_lin_quad_cS2
1063 prop_binCRWS_electron2017_bin13_sm_lin_quad_cS2
1064 prop_binCRWS_electron2016M_bin10_sm_lin_quad_cS2
1065 prop_binCRWS_muon2018_bin19_sm_lin_quad_cS2
1066 prop_binCRWS_electron2018_bin8_quad_cS2
1067 prop_binSR_muon2016M_bin16_sm_lin_quad_cS2
1068 prop_binCRF_muon2016M_bin17_quad_cS2
1069 prop_binCRF_muon2016M_bin18_quad_cS2
1070 prop_binCRWS_muon2017_bin19_quad_cS2
1071 prop_binSR_electron2017_bin6_quad_cS2
1072 prop_binCRWS_electron2017_bin15_sm_lin_quad_cS2
1073 prop_binCRWS_electron2018_bin8_sm_lin_quad_cS2
1074 prop_binCRWS_muon2016M_bin13_sm_lin_quad_cS2
1075 prop_binCRWS_electron2017_bin14_sm_lin_quad_cS2
1076 prop_binCRF_muon2016M_bin17_sm_lin_quad_cS2
1077 prop_binSR_muon2016M_bin16_quad_cS2
1078 prop_binCRF_electron2016M_bin15_quad_cS2
1079 prop_binSR_electron2016M_bin11_sm_lin_quad_cS2
1080 prop_binCRWS_electron2017_bin14_quad_cS2
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2



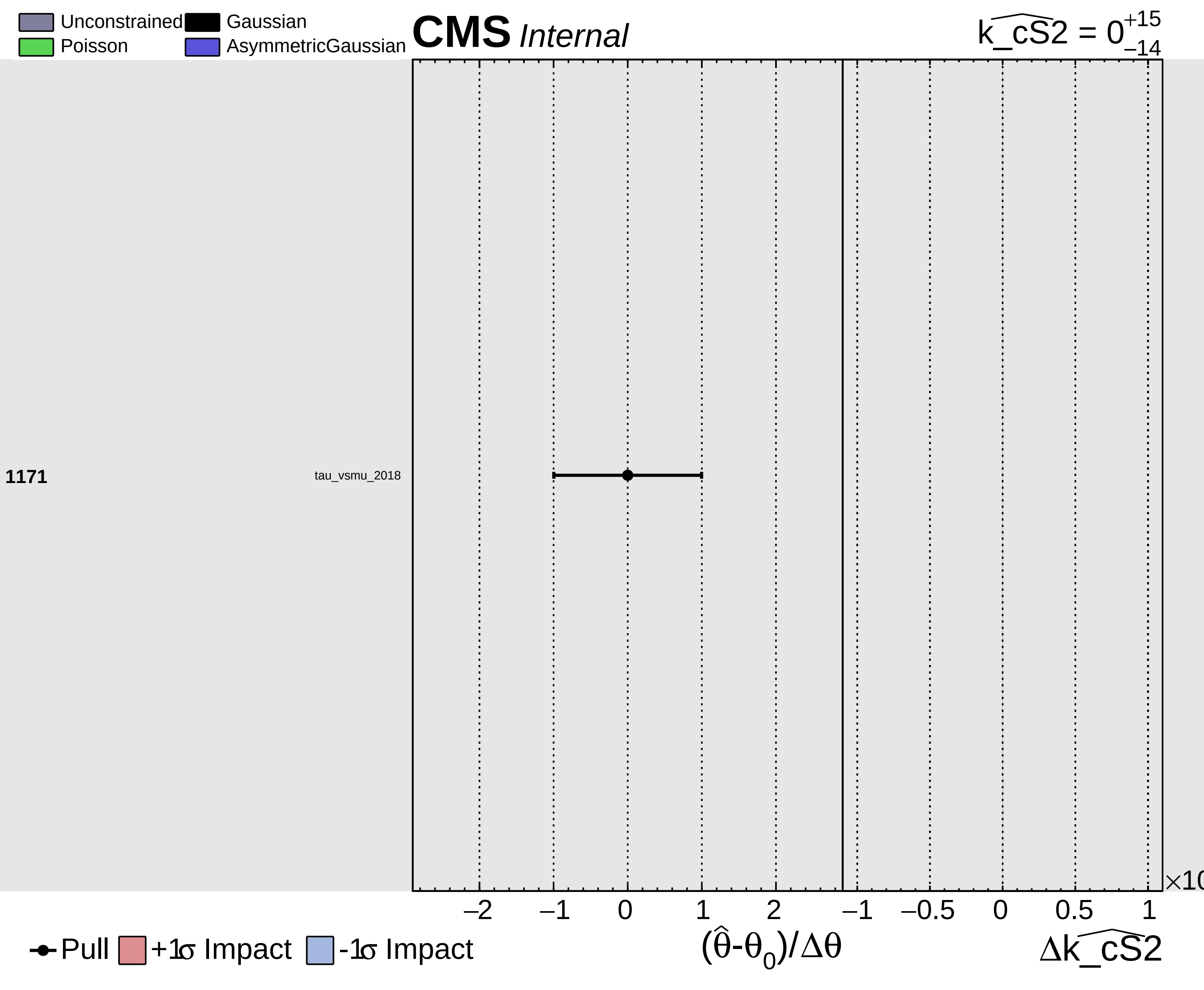
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
1081 prop_binSR_muon2017_bin11_quad_cS2
1082 prop_binCRF_muon2016M_bin19_sm_lin_quad_cS2
1083 prop_binCRF_muon2017_bin9_sm_lin_quad_cS2
1084 prop_binCRWS_muon2016M_bin11_sm_lin_quad_cS2
1085 prop_binCRF_muon2016M_bin15_sm_lin_quad_cS2
1086 prop_binCRF_electron2018_bin14_quad_cS2
1087 prop_binCRWS_muon2016M_bin16_quad_cS2
1088 prop_binSR_muon2016M_bin19_quad_cS2
1089 prop_binSR_muon2016M_bin18_quad_cS2
1090 prop_binSR_muon2016M_bin19_sm_lin_quad_cS2
1091 prop_binSR_electron2017_bin6_sm_lin_quad_cS2
1092 prop_binCRF_muon2016M_bin13_sm_lin_quad_cS2
1093 prop_binSR_muon2017_bin15_sm_lin_quad_cS2
1094 prop_binSR_electron2018_bin15_sm_lin_quad_cS2
1095 prop_binSR_muon2016M_bin18_sm_lin_quad_cS2
1096 prop_binSR_muon2017_bin11_sm_lin_quad_cS2
1097 prop_binCRF_muon2017_bin13_sm_lin_quad_cS2
1098 prop_binSR_electron2017_bin15_sm_lin_quad_cS2
1099 prop_binSR_muon2017_bin20_sm_lin_quad_cS2
1100 FES
1101 VBS_2016M
1102 VBS_2017
1103 VBS_2018
1104 prop_binCRF_electron2016M_bin11_sm_lin_quad_cS2
1105 prop_binCRF_electron2016M_bin12_sm_lin_quad_cS2
1106 prop_binCRF_electron2016M_bin15_sm_lin_quad_cS2
1107 prop_binCRF_electron2016M_bin16_quad_cS2
1108 prop_binCRF_electron2016M_bin16_sm_lin_quad_cS2
1109 prop_binCRF_electron2016M_bin18_quad_cS2
1110 prop_binCRF_electron2016M_bin19_sm_lin_quad_cS2
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
1111 prop_binCRF_electron2017_bin16_quad_cS2
1112 prop_binCRF_electron2017_bin16_sm_lin_quad_cS2
1113 prop_binCRF_electron2017_bin18_sm_lin_quad_cS2
1114 prop_binCRF_electron2018_bin14_sm_lin_quad_cS2
1115 prop_binCRF_electron2018_bin18_quad_cS2
1116 prop_binCRF_electron2018_bin18_sm_lin_quad_cS2
1117 prop_binCRF_electron2018_bin6_quad_cS2
1118 prop_binCRF_muon2016M_bin12_quad_cS2
1119 prop_binCRF_muon2016M_bin12_sm_lin_quad_cS2
1120 prop_binCRF_muon2016M_bin13_quad_cS2
1121 prop_binCRF_muon2016M_bin15_quad_cS2
1122 prop_binCRF_muon2016M_bin18_sm_lin_quad_cS2
1123 prop_binCRF_muon2016M_bin19_quad_cS2
1124 prop_binCRF_muon2016M_bin8_quad_cS2
1125 prop_binCRF_muon2017_bin13_quad_cS2
1126 prop_binCRF_muon2018_bin18_quad_cS2
1127 prop_binCRF_muon2018_bin18_sm_lin_quad_cS2
1128 prop_binCRWS_electron2016M_bin11_quad_cS2
1129 prop_binCRWS_electron2016M_bin11_sm_lin_quad_cS2
1130 prop_binCRWS_electron2016M_bin12_quad_cS2
1131 prop_binCRWS_electron2016M_bin12_sm_lin_quad_cS2
1132 prop_binCRWS_electron2016M_bin16_quad_cS2
1133 prop_binCRWS_electron2016M_bin16_sm
1134 prop_binCRWS_electron2016M_bin16_sm_lin_quad_cS2
1135 prop_binCRWS_electron2016M_bin6_sm_lin_quad_cS2
1136 prop_binCRWS_electron2017_bin10_sm_lin_quad_cS2
1137 prop_binCRWS_electron2017_bin13_quad_cS2
1138 prop_binCRWS_electron2017_bin15_quad_cS2
1139 prop_binCRWS_electron2018_bin8_sm
1140 prop_binCRWS_muon2016M_bin16_sm
×10
−2
−1
0
1
2
−1
−0.5
0
0.5
1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 0 +15 −14
1141 prop_binCRWS_muon2016M_bin16_sm_lin_quad_cS2
1142 prop_binCRWS_muon2016M_bin5_quad_cS2
1143 prop_binCRWS_muon2017_bin19_sm
1144 prop_binCRWS_muon2017_bin19_sm_lin_quad_cS2
1145 prop_binCRWS_muon2018_bin13_quad_cS2
1146 prop_binCRWS_muon2018_bin13_sm_lin_quad_cS2
1147 prop_binCRWS_muon2018_bin16_quad_cS2
1148 prop_binCRWS_muon2018_bin19_quad_cS2
1149 prop_binCRWS_muon2018_bin19_sm
1150 prop_binCRWS_muon2018_bin5_quad_cS2
1151 prop_binCRWS_muon2018_bin5_sm
1152 prop_binCRWS_muon2018_bin5_sm_lin_quad_cS2
1153 prop_binSR_electron2016M_bin11_quad_cS2
1154 prop_binSR_electron2016M_bin14_quad_cS2
1155 prop_binSR_electron2016M_bin14_sm_lin_quad_cS2
1156 prop_binSR_electron2016M_bin18_quad_cS2
1157 prop_binSR_electron2016M_bin18_sm_lin_quad_cS2
1158 prop_binSR_electron2017_bin14_quad_cS2
1159 prop_binSR_electron2017_bin14_sm_lin_quad_cS2
1160 prop_binSR_electron2017_bin15_quad_cS2
1161 prop_binSR_electron2018_bin15_quad_cS2
1162 prop_binSR_muon2017_bin15_quad_cS2
1163 prop_binSR_muon2018_bin11_quad_cS2
1164 prop_binSR_muon2018_bin11_sm_lin_quad_cS2
1165 r
1.0 +0 −0
1166 tau_vsele_2016M
1167 tau_vsele_2017
1168 tau_vsele_2018
1169 tau_vsmu_2016M
1170 tau_vsmu_2017
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 0^{+15}_{-14}$
1171
tau_vsmu_2018
$\times 10$
–2 –1 0 1 2
–1 –0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$